

Master Thesis

Fact Verification in Case Law Documents: An Industry Use Case

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Abstract

Automated fact verification is the task of determining whether a claim is supported or contradicted by a source. This has become an important task in natural language processing, and is increasingly applied to specialized domains such as case law. This thesis addresses fact verification in international case law documents, in collaboration with the International Bureau of Fiscal Documentation (IBFD), and examines how well a lighter-weight approach can perform this task compared to an LLM-based system. Existing legal fact verification work relies on large language models or domain-adapted transformers, with little systematic study of how well cheaper, more transparent approaches can perform on the same task. To study this, the thesis evaluates a lighter-weight approach that scores facts against source candidates using string-based similarity, embedding-based similarity, paraphrase detection, and natural language inference, and compares each against IBFD's LLM-based baseline. Using a dataset of 899 atomic facts generated from the summaries of 14 case law documents, each approach was evaluated at sentence-level and chunk-level, with an error analysis of the false positive cases (Contradicted class facts labeled as Verified) each produced. The lighter-weight approach did not match the baseline's macro-F1 score of 0.75, but chunk-level natural language inference reached a macro-F1 score of 0.64. The analysis showed that the main limitation lies in the structure of the task: Contradicted class facts differ slightly from otherwise near-identical source sentences, and contain a subset of irregular facts, therefore, challenging the performance of every approach, including the LLM baseline. The thesis offers IBFD a concrete basis for weighing fact verification performance against computational cost.

Declaration of Authorship

I, Bardia Kashani, declare that this thesis, titled *Fact Verification in Case Law Documents: An Industry Use Case* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: 06/08/2026

Signed: Bardia.K

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Chapter 1

Introduction

1.1 Motivation

Automated fact verification has become an important task in natural language processing (NLP). While general fact verification often targets public misinformation, specialized domains face a different challenge. In fields such as the legal domain, publishers must ensure that every analysis and summary they produce is completely faithful to the original source text, so they would not intentionally or unintentionally provide incorrect information to their audience. In case of providing misinformation, it causes severe professional and financial consequences for the publisher.

This thesis examines fact verification from case law documents in collaboration with the International Bureau of Fiscal Documentation (IBFD). Case law is a category of legal domain documents constituting a body of law made up of the interpretation and application of the law as established by judges and courts. Each case law document records the reasoning, evidence, and decisions in a specific legal dispute, and the ruling it contains has direct authority for the interpretation of law in subsequent cases. As a category of legal text, case law is characterized by formal vocabulary, numerous referencing between different cases and prior judgments, and a structured style designed to minimize ambiguity (Ariai et al., 2025). While this level of precision is necessary for making these documents authoritative, it also puts them at risk of providing wrong information with even minor alterations in elements such as numerical values. Verifying such claims is therefore a non-trivial task and requires close attention to detail at the level of individual claims (facts).

IBFD is an international tax-law publisher and provides various services for its users, including producing and curating case law content for legal professionals worldwide. As part of their work, IBFD must check and verify the information mentioned in the case law summaries they create, before providing them to their audience. Performing this verification task manually by in-house experts is time-consuming and expensive.

To solve this problem, IBFD's experts developed an LLM-based approach to perform the fact verification that compares facts generated from the case law summaries against source case law documents. Although powerful, this approach has certain drawbacks. LLMs are prone to hallucinations and they may produce outputs that sound fluent and plausible, yet they are factually incorrect or the information they provide is not supported by any source, a limitation that has been widely mentioned and studied across natural language generation tasks (Ji et al., 2022). These risks become more severe in the legal domain, where more formal vocabulary and syntax structure with

sensitive data make LLMs more susceptible to hallucinations (Ariai et al., 2025). These limitations have motivated the interest of this thesis in a lighter-weight fact verification approach that prioritizes computational efficiency, cost-effectiveness for IBFD, and transparency through access to the underlying scores. Because every approach provides the specific source candidate the fact was compared with, plus the pair's score, the performance and the reason behind the decision can be directly examined.

This thesis investigates how such a lighter-weight approach – using string-based comparison, embedding similarity, paraphrase detection, and natural language inference (NLI) – can perform for fact verification compared to the IBFD LLM-based fact verifier when applied to case law documents. The work was conducted in collaboration with IBFD, whose internal evaluation provided the dataset used in this thesis. The dataset consists of 14 case law projects and 899 atomic facts extracted from summaries; the summaries have been authenticated by an in-house expert. An atomic fact is a short statement that contains only one piece of information (Nenkova and Passonneau, 2004; Zhang and Bansal, 2021; Liu et al., 2023). For example, the sentence "In the 2019 award, it was decided that Spain breached the agreement and awarded the claimant 100 million euros" can be separated into distinct atomic facts: `"The award was issued in 2019"`; `"It was decided that Spain breached the agreement"`; and `"The claimant was awarded 100 million euros."` Finally, each of these atomic facts is given one of the following three labels based on the information in the case law documents:

- **Verified**: The claim is supported according to the text in the document.
- **Contradicted**: The claim is denied according to the text in the document.
- **Insufficient Evidence**: The text is not informative enough to either verify or deny the claim.

For this thesis, the three labels are collapsed into a binary evaluation (Section 3.1.2)

1.2 Research Questions

To systematically evaluate the performance of different approaches, this thesis addresses the following main question:

- How do alternative lighter-weight approaches compare to IBFD LLM-based system performance for fact verification in case law documents?

Considering the utilization of the four mentioned approaches, their individual performance on fact verification is also important to explore and compare. Therefore, the following sub-questions are addressed:

- How do string-based edit distance methods perform for fact verification in case law documents?
- How do approaches based on embedding similarity perform for fact verification in case law documents?
- How do more semantically informed methods based on paraphrase detection and natural language inference, perform for fact verification in case law documents?

1.3 Summary of Results

Across the evaluated case law documents, the four lighter-weight approaches reach broadly similar performance at the sentence-level, with macro-F1 scores between 0.54 and 0.57. The clearest improvement comes from natural language inference when applied to multi-sentence chunk-level evaluation. The large cross-encoder NLI model reaches a macro-F1 of 0.64, the closest any lighter-weight configuration comes to the IBFD LLM-based baseline of 0.75. The error analysis (Chapter 5) shows that the main obstacle is not the choice of scoring approach but partly the structure of the data, where contradicted facts are typically minimally edited paraphrases of the source text, and partly a capability gap between the lighter-weight approaches and the LLM-based system.

1.4 Outline

The thesis is outlined as follows: Chapter 2 provides the related work on the past and current research on fact verification approaches. Chapter 3 elaborates on the detailed methodology for performing fact verification, how each approach examines and evaluates every fact-candidate pair, and how the scoring is done for each pair. Chapter 4 reveals the results produced by each approach and provides a comparison. Chapter 5 further reports the error analysis carried out across all utilized approaches, identifying systematic patterns of misclassifications. Chapter 6 discusses the limitations of this project, along with potential improvements and new directions for future research, and draws a conclusion.

Chapter 2

Related Work

2.1 Automated Fact Verification

Fact Verification is the task of determining, given a claim and a body of evidence retrieved from an authentic knowledge source, whether the evidence supports, refutes, or the data provided is insufficient to either verify or deny the claim. In recent years, with the advancement of technology, the volume of the textual context has increased across digital platforms, most of which have not been verified. The recognition of this occurrence was captured in 2009, when the Pulitzer Prize for National Reporting was awarded to PolitiFact[1], a fact verification system that managed to systematically examine the political claims during the United States presidential political campaign and verified the correct claims (Quelle and Bovet, 2024). Since 2009, the fact verification procedure has changed and evolved from a journalistic perspective into a more research-related area within Natural Language Processing (NLP). A growing need for automated fact verification approaches has arisen due to the large amount of text produced today, as a result of the advances in technology and the practical impossibility of manually performing this work.

Reflecting this procedure, Rashkin et al. (2017) report in their study that the number of organized initiatives for fact verification nearly tripled from 2014 to 2017. Exposure to misinformation has a measurable cognitive effect on readers and can lead to a behavioral consequence: confusion, building false beliefs, mindsets, and ideas, even if the publisher retracts the original claim (Morris et al., 2020; Porter and Wood, 2021). Having this in mind, the operational efficiency of developing such automated fact verification methods becomes necessary.

Vlachos and Riedel (2014) established the framework for automating fact verification by performing a retrieval-based verification on a smaller scale. The establishment of retrieval-based automated fact verification as a large-scale baseline computational task was defined by the FEVER dataset (Thorne et al., 2018), which basically structured verification as a three-step pipeline: for a given textual claim, the system must initially gather relevant evidence from a larger reference corpus, followed by classifying the relationship between the claim and the evidence as *Supported, Refuted* or *NotEnoughInfo*. FEVER contained 185,445 claims and managed to establish the three-label

[1]Prestigious annual award established by newspaper publisher Joseph Pulitzer and administered by Columbia University since 1917 for excellence in journalism, literature, and music. It features 23 categories, including 15 for journalism—where the Public Service winner receives a gold medal—and categories for fiction, drama, and history

framework that has been globally accepted and adopted across various fact verification benchmarks.

From here, the methods used in automated fact verification have followed the overall development of NLP. Trancoso et al. (2024) describes this progression across three main stages: up to 1980s, NLP systems were primarily dependent on heuristic rules; from late 1980s onward, statistical machine learning techniques were adopted, allowing system to derive rules from data itself; and from 2010s onward, deep learning and representation learning became dominant, with word embeddings capturing semantic connection and transformer models (Vaswani et al., 2017) providing the opportunity for parallel computation over long sequences. An important breakthrough was the introduction of self-attention by Vaswani et al. (2017), making it possible for the models to simultaneously weight contextual relevance of every token to every other token, enabling later development in pre-trained language models such as BERT (Devlin et al., 2019).

Many modern fact verification pipelines are processed by Large Language Models (LLMs), which can be prompted to assess claims against retrieved evidence. As established by Lewis et al. (2020), where they introduced their foundational Retrieval-Augmented Generation (RAG) framework, RAG incorporates a differentiable retrieval mechanism allowing a sequence-to-sequence model to benefit from an external knowledge base as additional context. This architecture ensures that the final output is generated by using reliable, non-parametric source documents, therefore turning an unconstrained generation task into a factual, evidence-based verification pipeline. However, LLM-based verifiers have notable drawbacks. First, they are computationally expensive to run and require substantial memory and processing resources for deployment (Li et al., 2022). Second, their reasoning is difficult to interpret due to off-the-shelf models struggling with capturing the full complex structure and fluctuating interpretations regarding legal languages (Padiu et al., 2024). Finally, they are highly susceptible to hallucination, with at least 58% rate in specific legal domain queries (Dahl et al., 2024). Moreover, Bender et al. (2021, p. 611) notes that LLMs are trained for "predicting the likelihood of a token, given either its preceding context or its surrounding context", which makes them more probabilistic predictors rather than truly logical verifiers. These limitations have rooted the interest in having a lighter-weight, feasible approaches, particularly in specific domains where transparency and computational efficiency are required.

The fact verification task is also highly sensitive to the domain it is being applied to. General-purpose systems trained on either Wikipedia or news data would, most likely, perform poorly on specialized text in the legal domain. The case law documents carry their own vocabulary, structural conventions, and reasoning patterns (Ariai et al., 2025). Therefore, the following section aims to narrow the focus to fact verification in the legal domain, where these challenges are mainly faced.

2.2 Fact Verification in the Legal Domain

Although general-domain fact verifications applied to benchmarks like FEVER or Wikipedia has produced robust methods, adapting them to a specific, specialized domain (e.g., biomedical or legal) is far from straightforward, since they contain technical vocabulary and unique structure. Texts specified for the legal domain exhibit a set of linguistic and structural properties that distinguish them from other domains, such as

news articles or daily conversational data, on which most NLP systems are trained and evaluated. Understanding these linguistic and structural properties is an essential step for having a fact verification system operable on legal documents.

As an example, Ariai et al. (2025) identified some defining characteristic of legal documents. First, the legal language used inside these documents is highly formal, contains specialised vocabulary, fixed syntactic patterns, and numerous punctuation marks that are designed to provide clarification and minimize ambiguity. This point can be vital in case law documents because, as a category of legal domain, they rely heavily on precise, carefully written language, where minor wording differences can change the legal consequences (Ariai et al., 2025). Therefore, even minor variations, such as small changes in modifiers of numerical values, can carry a semantic weight which a model trained on general texts is not equipped to detect. These minor changes can result in direct consequences for any fact verification system aiming to compare the claims to their source text.

As for the second characteristic, legal domain language relies heavily on specialised and unique vocabulary, including terms that have special meaning in the legal domain, archaic expressions rarely found in everyday text, and phrases embedded in legal tradition (Ariai et al., 2025). Trancoso et al. (2024) supported this claim, noting that legal document texts contain particular syntax and semantics in domain-specific information, plus a tendency towards using long sentences. All these properties affect the performance of general models and support the need for moving towards domain-specific model adaptation, as the general lexical and syntactic assumptions encoded in general models often misrepresent legal vocabulary and structure.

Third characteristic suggested by Ariai et al. (2025) is that legal documents, especially case law documents, are heavily intertextual. Due to their content, the information they need to cover, and the examples they need to provide, they need to bring references, regulations, prior judgments, and other document citations. Therefore, on many occasions, interpreting a single passage needs access to a broader legal framework. In the context of case law fact verification, this intertextuality can complicate the procedure, as the information required to capture whether a fact is mentioned correctly in the document or not, is dependent on multiple sections of the document.

In addition, the structural format of legal documents adds further difficulty for fact verification. Ariai et al. (2025) mention that legal documents are normally provided as non-machine-readable PDFs, which heavily complicates text extraction and processing. Long document lengths, footnotes, digits, and unique punctuation cause problems in capturing a clean, sentence-segmented input an NLP pipeline prefers.

The complexity of the legal domain has motivated further work on domain-adapted language models and benchmark datasets. Chalkidis et al. (2020) introduced LEGAL-BERT, a family of BERT variants pre-trained on legal corpora, demonstrating that domain-adapted model manage to outperform the general models' counterparts on legal NLP tasks. Subsequent benchmarks such as LexGLUE (Chalkidis et al., 2022) aggregated multiple legal NLP tasks into a single evaluation covering multi-label classification, multi-class classification, and multiple-choice question answering. However, as Ariai et al. (2025) observed in their review of 131 studies in legal NLP, development of specialised legal corpora remains underdeveloped, relative to other better-studied tasks, such as summarization and judgment prediction. The verification of factual claims within case law documents has therefore received, comparatively, less attention in recent studies.

A more relevant work has been carried out by Holzenberger et al. (2020), who introduced the Statutory Reasoning Assessment (SARA) dataset, which framed legal verification as a procedure of the Natural Language Inference (NLI) task within the United States tax law. SARA performs by matching pairs of facts about hypothetical taxpayers with defined provisions and checks whether the provision entails or contradicts the claim in a binary classification framework. Koreeda and Manning (2021) managed to expand upon this framing to document-level NLI in legal contracts with the help of the ContractNLI dataset. Unlike SARA's binary approach, here ContractNLI classifies each hypothesis against an entire contract, while using the full three-class NLI framing (entailment, contradiction, neutral). Together, these datasets illustrated that legal verification can be cast as a textual entailment problem, which motivates the use of an NLI component for legal domain tasks.

Despite the progress, a clear gap still remains. Existing legal fact verification relies heavily on LLMs or domain-adapted transformer models, while little systematic attention given to whether there are lighter-weight approaches to perform the same task and how well they can perform.

2.3 String-Based Comparison Using Edit Distance

Addressing the mentioned gap requires examining how well lighter-weight approaches can perform. The simplest method of applying a similarity comparison is exact string match. In order to move a step further from the exact matching while remaining computationally inexpensive, an approximate string matching can be applied, a set of techniques originally surveyed by Navarro (2001). Approximate string matching contains several distance comparison approaches, including Jaro, Jaro-Winkler distance, and the Levenshtein distance. Among these approaches, the Levenshtein distance has became the most widely adopted metric for character-level similarity (Levenshtein, 1966). Levenshtein distance is defined by the minimum number of single character changes required to transform one string into another, with three methods of insertion, deletion, and substitution. Coates and Breitinger (2023) mention that Levenshtein distance provides the opportunity for both mathematical precision and intuitive meaning for human readers, so it provides an explanation and an output which is intuitive to understand, compared to, for example, embedding output that is opaque and impossible for a human to intuitively grasp. Therefore, Levenshtein approach is a suitable option for a verification task where the matches require to be interpretable.

The Levenshtein distance has a computational limitation: it performs poorly because its complexity is quadratic in the length of the two strings being compared, which makes naive and basic implementations impractical for very large documents Coates and Breitinger (2023). To address these computational bottlenecks, the Levenshtein algorithm has been optimized for creating more advanced libraries. One such library is RapidFuzz **?**, an open-source Python library that provides implementations of Levenshtein-related similarity comparison metrics. While being methodologically flexible, RapidFuzz is also well suited to the large scale text comparison. Elmobark (2025) provides a comparative benchmark of the five most commonly used Python text-matching libraries across 50,000 multilingual test cases, and indicates that RapidFuzz achieves the highest processing speed among others. In addition, for scalability testing, RapidFuzz and Levenshtein both maintain a near-linear performance up to 100,000 text pairs, while other libraries' performance drops beyond 50,000.

Using edit-distance based methods on noisy legal text has been examined by the prior research. Elmobark (2025) also reports that RapidFuzz and Levenshtein both achieve 98.5% accuracy on missing-character errors, while RapidFuzz handles extra-character errors with 98.2% accuracy. Hladek et al. (2017) examine the learned string distance, while focusing on smoothing to the Optical Character Recognition (OCR) output of the texts, and prove that edit-distance based approaches are actually effective for correcting the error resulted from OCR, including the domain containing specialized conventions. Recently, McCammon (2025) evaluated the use of fuzzy matching for the retrieval task over noisy data, mentioning that approximate matching outperforms the exact matching when the source data contains unbounded variation. Although neither studies address fact verification directly, they do illustrate a more diverse principle that edit-distance approaches manage to recover meaningful matches in textual settings where either OCR or other methods causing noise are present.

While proved to be applicable, string-based comparison methods are surface-level. They cannot recognize the similarity between two sentences when they are expressing the same logic, but with different vocabulary, nor can they identify the matches between two sentences where one is syntactically and structurally different from the other. These limitations motivate moving towards a second matching step based on semantic embeddings similarity.

2.4 Embedding Similarity

String-based methods capture character-level similarity but cannot capture semantic relation. This limitation has motivated a shift in the literature toward embedding-based approaches. Standard embedding similarity approaches use models like BERT (Devlin et al., 2019) to encode texts into dense vectors, comparing them via cosine similarity: the basic and standard comparison function. This method of comparing similarity makes it especially suitable for comparing embeddings of sentences with different lengths (Reimers and Gurevych, 2019). However, BERT with *cross-encoder* comparison scales poorly, because comparing all possible pairs in a long document requires a separate path for each claim-sentence pair, making it computationally expensive for document-level fact verification (Reimers and Gurevych, 2019). To solve this limitation, Reimers and Gurevych (2019) introduced Sentence-BERT (SBERT), a fine-tuned version of BERT that uses Siamese neural network architecture (Bromley et al., 1993) to directly compare fixed-size sentence embeddings with cosine similarity. This bi-encoder design leads to individual encoding for each sentence, therefore lowering the cost of large size similarity comparison while preserving the semantic quality of the transformer representations.

Applying embedding-based similarity to legal domain texts has received much attention in recent works. Fan et al. (2024) proposed an ensemble-learning architecture for matching the context of legal cases. A core component of their system was a representation-based sub-network trained with contrastive learning to put semantically similar cases closer in the embedding space. Their results, based on *CAIL2019-SCM* benchmark, illustrated that these dense embeddings are highly effective at capturing the semantic distribution of legal texts, serving as a suitable approach for large-scale legal similarity comparison.

Donnelly and Roegiest (2025) examined whether ready-made embedding models are capable of capturing semantic similarity between legal interpretation of contract

clauses. They utilized five different embedding models, including LegalBERT and OpenAI's *text-embedding-3-large*, and they compared cosine similarity between an original, authentic legal clause and two variants: a smaller, edited version of the clause to act as the negated version that reverses the clause meaning, and a fully rephrased version that contains the same meaning but expresses it in a different syntactic structure. Based on their conclusion, across all models and similarity comparisons tested, the negated variant is consistently placed as more similar to the original than the rephrased variant, representing the difficulty of capturing the similarity when one has been paraphrased. The authors attributed this to embeddings that remain "heavily grounded in the lexical and syntactical structure of the text rather than trying to represent higher-order meaning" (Donnelly and Roegiest, 2025, p. 402). So, they concluded that ready-made, off-the-shelf embedding similarity models should be used with caution in the legal domain.

2.5 Paraphrase Detection and Natural Language Inference

While string-based comparison and embedding similarity approaches manage to capture surface or topical similarities, they might fail to recognize them if there are alterations in the structure or wording. For logical similarity, the literature examines two more semantically-informed approaches, grounded in capturing the semantic relationship between each claim-sentence pair: paraphrase detection and Natural Language Inference.

2.5.1 Paraphrase Detection

Paraphrase detection is the task of determining whether two sentences express the same claim and meaning, despite having different surface form and structure. It is typically performed as a binary classification problem (Michail et al., 2025). However, paraphrase detection has its limitations. Michail et al. (2025) introduce PARAPHRASUS, a multi-dataset benchmark for paraphrase detection that spans 10 test sets across three main challenges: classification, minimization of false positives, and maximization of true paraphrase detection. By looking at their findings, no current models, including large language models with carefully engineered prompts, achieve consistent strong performance across all three criteria. In addition, they report that state-of-the-art models can also be misled by simple linguistic variations such as changing the verb to a passive form. These results support the position adopted in this thesis that paraphrase detection should be treated as one approach among other approaches, rather than being considered as a standalone solution.

2.5.2 Natural Language Inference

While paraphrase detection asks whether two sentences express the same claim in a different format, Natural language inference examines a more in-depth question about their logical relationship. Natural language inference (NLI), also called recognizing textual entailment (RTE), determines whether a given a *premise* entails, contradicts, or is neutral towards a *hypothesis*.

NLI task was explicitly studied by Bowman et al. (2015) while working on the Stanford Natural Language Inference (SNLI) corpus, which provided around 570,000

human-annotated sentence pairs labeled with either entailment, contradiction, or neutral. SNLI established the three categories that have been used across later NLI resources and became the base for the FEVER fact extraction framework discussed in section 2.1.

The NLI framing has been previously applied in the legal context domain by, among others, Holzenberger et al. (2020) by introducing SARA for statutory reasoning and Koreeda and Manning (2021) for document-level inference in contract evaluation (both discussed in Section 2.2). As Reimers and Gurevych (2019) mentioned, NLI models are typically implemented as cross-encoders, in which both premise and hypothesis are passed through a transformer; therefore, comparison is done on the pair rather than individually. This architecture captures fine-grained relationships between premise and hypothesis tokens. However, the computational time to have an individual comparison for every claim-sentence pair is significant. DeBERTa (He et al., 2021b) refines BERT and RoBERTa with a disentangled attention mechanism, which enables it to represent each token's content and relative position with separate vectors and then compute attention over both. This improves the performance including a +0.9% gain over RoBERTa-Large on the MNLI natural language inference task (He et al., 2021b), making DeBERTa-based cross-encoders a strong choice for NLI.

Although NLI offers an in-depth evaluation of semantic and logical structure, which embedding similarity does not, it still has certain limitations in the legal domain. Koreeda and Manning (2021) illustrate that *negation by exception*, a linguistic phenomenon common in contracts in which a clause is qualified by a direct subsequent exception, noticeably degrades NLI performance even for transformer-based models. Their research showed that facing such negations in the dataset can reduce the contradicted class F1 score. Case law documents exhibit similar patterns, so this limitation is directly relevant to the present task, because NLI may lose its advantage on facts whose verification depends on subtle or context-dependent information rather than a clear sentence-level claim in the source text.

Moreover, Laban et al. (2022) shows that restricting NLI to a single sentence for comparison can limit its effectiveness when the evidence for summarized sentences is spread across multiple source sentences. They demonstrate that expanding the evaluation granularity to multi-sentence chunks provides the NLI model with the required context and improves performance.

2.6 Summary

This chapter reviewed the previous works on four approaches for fact verification in the legal domain. String-based comparison approach (Levenshtein and RapidFuzz) is computationally efficient and interpretable but fail at recognizing meaning-preserved rewording structure (Coates and Breitinger, 2023). Embedding similarity captures topical and semantic overlap, but is insensitive to deeper logical structure (Donnelly and Roegiest, 2025). Paraphrase detection captures sentences that convey the same meaning while having different structure, but is less reliable when lexical overlap is present (Zhang et al., 2019; Michail et al., 2025). The natural language inference framework manages to capture entailment and contradiction explicitly, but can be affected by legal domain specific phenomena such as negation by exception (Koreeda and Manning, 2021). These complementary strengths and limitations have motivated evaluating the four lighter-weight scoring approaches in the case law fact verification

task, and comparing them to the performance of an LLM in the present study.

To address the gap between these methods, the main research question of this thesis asks how lighter-weight approaches compare against the LLM-based system. Accordingly, three sub-questions systematically examine the performance of the specific approaches evaluated in this study: string-based comparison (SQ1), embedding similarity (SQ2), and more semantically informed approaches based on paraphrase detection and natural language inference (SQ3).

Chapter 3

Methodology

This chapter describes the steps taken during the investigation of the main research question of this thesis: the extent to which lighter-weight scoring approaches can perform in automated fact verification in case law documents, compared to a large language model (LLM) based system. The task is framed as a binary classification: given an atomic fact (Section 1.1) and its source document, the aim is to predict whether the information inside the document supports the fact (Verified) or rejects it (Contradicted).

The following sections are structured to provide a comprehensive sequential breakdown of the pipeline's architecture, providing a transparent and methodologically reproducible approach. To illustrate this procedure, Section 3.1 introduces the IBFD dataset and the procedure of achieving it. Section 3.2 elaborates on the overall pipeline structure designed for this thesis; Section 3.3 discusses the steps taken for text extraction from the source documents and having final, ready-to-implement text files; subsequently, Sections 3.4 through 3.7 elaborate on the specific steps for filtering and scoring approaches in this study. Furthermore, Section 3.8 presents the IBFD LLM-based results as an upper-bound baseline; Section 3.9 discusses chunk creation and comparison. Finally, the chapter concludes by defining the evaluation metrics and mapping the steps to the research questions (Section 3.10).

3.1 Data

The dataset used in this thesis was provided by IBFD. It is the internal set utilized for evaluating their own LLM-based fact verification system. The dataset consisted of 14 case law projects from the IBFD case law collection, each containing one or more source PDFs (award documents), together with 899 atomic facts generated from the summaries of these documents. These case law projects and the internal company evaluation framework are strictly confidential. However, explicit permission has been given by IBFD to utilize and discuss the examples and cases mentioned in this thesis.

3.1.1 Award Documents and Summaries

The award documents are highly formal, dense, intertextual, and often contain footnotes, tables, charts, and complex formatting. They range from 30 pages to 900 pages. The specialized and low-frequency vocabulary characteristic of the legal text (Section 2.2) is reflected in these award documents' vocabulary. The number of unique words per document ranges from 1,600 to 17,500 (see Appendix A, Table A.1), indicating

the lexical diversity the approaches must handle. As these documents are large and complex, a third-party person from outside the IBFD summarized the award documents. To create a usable dataset containing both supported and unsupported claims, an IBFD expert introduced corrupted facts by doing small alterations on some of the sentences inside the summaries, for example, changing a date, a number, or a key term, so that they contradict the source document. As the next step, an LLM was used to extract atomic facts from this pool of correct and corrupted summary sentences, and an IBFD expert then assigned each fact a gold label (Section 3.1.3).

It should be noted that during this process, rarely any verbatim statement was quoted or copied from the award documents, and the information was paraphrased. The main implication of this paraphrasing is that the facts are rarely verbatim restatements of the source, so simple exact word-matching approaches would struggle to verify facts correctly. This characteristic of the dataset motivated the approaches evaluated in this thesis, which compare facts and candidates. This comparison occurs beyond the surface form wording, at levels of approximate string matching and semantically informed approaches such as embedding, paraphrase detection, and natural language inference to capture the similarities.

This method of generating Contradicted class facts, by making small alterations, creates a challenge for the scoring approaches. By this design, the manually corrupted facts are nearly identical to a genuine source sentence. Therefore, it can be hypothesized that approaches relying on a score indicating how similar a fact and a sentence are, will struggle to capture these corrupted facts, since they will produce a high score against the source sentence. This structural limitation is examined directly in the results and error analysis (Chapters 4 and 5).

3.1.2 Dataset Creation and Annotation

Each atomic fact was labeled as either Verified, Contradicted, or Insufficient Evidence: a fact is Verified when its claim is supported in the corresponding award document, Contradicted when the target fact is false according to the corresponding document, and labeled as Insufficient Evidence when the available information is not sufficient to either verify or deny the fact. This annotation was carried out by a single in-house domain expert at IBFD; therefore, the inter-annotator agreement (IAA) metric could not be calculated.

Class	Total Facts	Distribution (%)
Verified	747	83.1%
Contradicted	132	14.7%
Insufficient Evidence	20	2.2%
Total	**899**	**100.0%**

Table 3.1: Dataset statistics and class distribution.

Table 3.1 presents the class distribution. Out of 899 facts, 83% belong to the Verified class. The Contradicted class represents only 14.7% of the data, while only 2.2% is dedicated to Insufficient Evidence. This imbalanced distribution supports the focus on the macro-average F1 score as the main evaluation metric in Chapter 4. Considering the label distribution, it is essential to address a necessary modification made to this dataset

Document	Verified	Contradicted	Insufficient Evidence	Total
9REN v. Spain	84	7	2	93
Ampal v. Egypt	60	9	0	69
Burlington v. Ecuador	75	15	2	92
Duke Energy v. Ecuador	57	7	3	67
Enron v. Argentina	51	9	1	61
Feldman v. Mexico	42	5	2	49
JSW Solar v. Czech Republic	36	14	3	53
Lone Star v. Korea	45	2	2	49
Mamidoil v. Albania	33	17	0	50
Manolium-Processing v. Belarus	31	2	2	35
Paushok v. Mongolia	82	6	2	90
Perenco v. Ecuador	47	19	0	66
Tokios Tokelés v. Ukraine	49	12	1	62
Yukos Universal v. Russia	55	8	0	63
Total	**747**	**132**	**20**	**899**

Table 3.2: Per-document fact counts in the IBFD case law dataset.

prior to pipeline evaluation. Considering the extreme sparsity, capturing and evaluating the Insufficient Evidence cases as a third, independent class would distort macro-average evaluation metrics. Moreover, the current lighter-weight pipeline architecture is mainly designed as a binary verification system utilizing a singular classification threshold. Therefore, the 20 Insufficient Evidence instances were merged into the Contradicted class for the evaluation, resulting in a binary distribution of **747 Verified** and **152 Contradicted** facts.

As illustrated in Table 3.2, this imbalance is consistent across documents. Most facts extracted from the summaries are Verified and the distribution of all facts per document ranges from 35 to 93, with *9REN v. Spain* showing the largest gap between Verified and Contradicted.

3.1.3 Instance Representation

Each classification compares an atomic fact against its corresponding candidate in a specific award document, and each has received a human-annotated gold label. Regarding the dataset's granularity, these labels apply per-document. Therefore, the dataset illustrates whether a fact is Verified or Contradicted. As Table 3.3 shows two examples of the facts and labels provided as a JSON file, in addition to their source document and gold label.

Document	**Extracted Fact**	**Human Verification (Gold)**
9REN v. Spain	"The date of the celebration is 29 June 2019."	Contradicted
Ampal v. Egypt	"Breaches of international investment agreement were found"	Verified

Table 3.3: Examples of Instances inside the JSON

3.2 Pipeline Overview

The pipeline designed for this thesis first captures the extracted texts of the award documents and atomic facts generated from their summaries, and produces as output a binary label classification indicating whether a fact is verified or contradicted, according to the source document. Figure 3.1 presents the sequential stages of the pipeline designed in this study. First, the award documents undergo pre-processing and candidates for comparison are filtered (Section 3.3). Each scoring approach is elaborated in detail in Section 3.4 throughout Section 3.7.

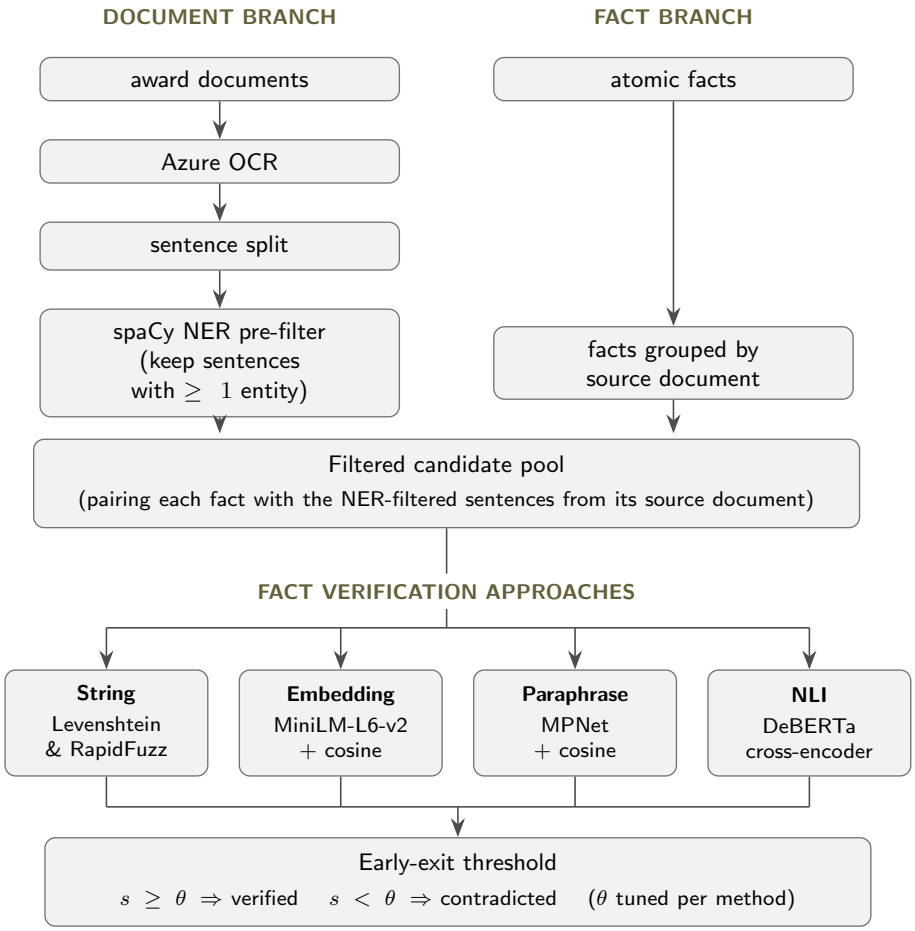


Figure 3.1: Pipeline architecture for fact verification in case law documents.

As illustrated in Figure 3.1, the verification architecture is designed based on two independent branches that converge into a unified fact-candidate pair comparison step. The document branch sequentially processes the raw award documents through optical character recognition (OCR) text extraction, followed by sentence segmentation, while the fact branch groups the atomic facts by their source document. The segmented sentences undergo a pre-processing step of using NER as a pre-filter and any sentence without at least one named entity is discarded, noticeably reducing the candidate pool. Each fact is then compared only against the remaining candidate sentences from its respective source document, leading to cleaner, less noisy comparisons. To evaluate the effectiveness of this step, the pipeline is executed both with and without the NER pre-filter and the overall performance is compared. The four distinct scoring approaches

performed are:

- **string-based edit distance**
- **dense semantic embedding similarity**
- **paraphrase detection**
- **cross-encoder natural language inference (NLI)**

Finally, each scoring approach produces a score for every fact-candidate pair. For each fact, the score is evaluated against a predefined threshold. If any candidate meets the threshold, the fact is considered Verified. If none of the candidates meet the threshold, the fact is considered Contradicted (see Listing 3.1). The candidate responsible for the decision is then retained for the error analysis in Chapter 5. As detailed in Section 3.10, multiple thresholds have been experimented.

```
IF S ≥ θ
    prediction → "Verified"
    BREAK
ELSE
    prediction → "Contradicted"
```

Listing 3.1: Threshold (θ) decision logic for fact verification based on the final score S.

3.3 Pre-Processing

As noted in Section 3.1, the award documents were provided as PDFs. These documents are therefore designed as fixed-layout visual representations rather than being structured as text containers, which makes text extraction more difficult than for other document types. A basic text-extraction library often scrambles the reading order, especially for legal documents, which usually contain multi-column tables, charts, and footnotes. As the sentence-level comparison requires cleanly separated sentences, OCR was used to convert award documents into iterable texts. This thesis uses Azure Document Intelligence [1], which applies AI models to perform OCR and extract the text from award documents, transforming them into usable data. Because Azure operates as a serverless API endpoint, the extraction procedure functions as a "black box", requiring the pipeline to accept the extracted texts from the award documents as they have been provided. The extracted text was then segmented into sentences: although different methods such as Regular Expression (regex) were examined, the segmentation was eventually performed based on the newline character (\n). Moreover, out of 14 projects provided by IBFD, 6 contained more than one award document related to a single summary. For these projects, the OCR output of each award document was then merged into a single file before segmentation, so that every fact could be paired against a single source. This results in one extracted text per project and 14 award document texts in total for evaluation.

The described procedure produced approximately 172,000 sentences, distributed per document as shown in Table 3.4. While each fact is compared only against the

[1] https://learn.microsoft.com/en-us/azure/ai-services/document-intelligence/overview

Case Name (Shortened)	Number of Sentences
Perenco v. Ecuador	35,086
Enron v. Argentina	991
Feldman v. Mexico	2,774
9REN Holding v. Spain	6,640
Duke Energy v. Ecuador	5,805
Lone Star v. Korea	21,240
Mamidoil v. Albania	6,579
Manolium-Processing v. Belarus	8,980
Paushok v. Mongolia	8,309
Tokios Tokelés v. Ukraine	3,267
JSW Solar v. Czech Republic	5,399
Yukos Universal v. Russia	35,205
Ampal v. Egypt	9,369
Burlington v. Ecuador	22,586

Table 3.4: Total Number of Extracted Sentences per Case File

sentences of its related document, this still means comparing a fact against thousands of sentences for the majority of cases. This search is computationally expensive for the scoring approaches and introduces noise from unrelated sentences. Considering that almost 82% of the facts contain named entities (739 out of 899), an optional NER pre-filtering step was applied by using spaCy, retaining only the sentences that contain at least one named entity. Two spaCy models were evaluated for this step: the CNN-based large and transformer models. While both models retained a comparable number of sentences (122,995 and 119,441 respectively), the transformer model was adopted because it produced a slightly smaller candidate pool. By discarding approximately 52,000 entity-less sentences, this pre-filtering reduces the number of fact–candidate pairs passed to the more computationally expensive scoring approaches; most notably the cross-encoder NLI model. Moreover, to empirically evaluate the effect of this step, this task is evaluated both with and without the NER pre-filter, with the comparative results reported in Chapter 4.

3.4 String-Based Comparison Scoring

The core idea is to compare each fact against each sentence in the document and decide whether the fact has been extracted from this sentence, and whether it verifies the fact or not. Initially, an exact match comparison was implemented to have a baseline understanding of how the facts are structured. This analysis revealed that out of 899 facts, only 2 were included as an exact sub-string inside the candidate sentences. This indicates that the facts were paraphrased and to find the correct fact-candidate pair, other approaches should be utilized; therefore, the exact string approach was excluded.

3.4.1 Levenshtein Edit Distance

One approach was to compare each pair's strings and examine how different they are from each other; Levenshtein distance examination was chosen for this approach (Levenshtein, 1966). Levenshtein distance is a character-level measure of the similarity between two strings. To implement Levenshtein as a similarity comparison approach in this thesis, Levenshtein Python library has been utilized. This library contains several

methods for comparing and matching strings. A brief explanation of the methods used in this thesis is provided in Table 3.5.

Method	Description
`RATIO`	Calculates the minimum number of changes required to change one sequence into the other, then outputs a normalized similarity score.
`SET_RATIO`	The best match between any strings in the first set and the second set (passed as sequences); the order does not matter here.
`SEQ_RATIO`	Computes the similarity ratio of two sequences of strings. This is like `ratio()`, but for string sequences.

Table 3.5: Description of Utilized Levenshtein Library Matching Methods

3.4.2 Token-Aware Edit Distance with RapidFuzz

To compute edit distance similarity more efficiently, while accounting for word order and structural variation, RapidFuzz library (Banchmann, 2021) has also been used for this approach. Building upon Levenshtein ratio as the foundational edit distance algorithm, RapidFuzz is a fast performing string-matching library extending traditional edit distance into token-aware comparison. While RapidFuzz internally performs the evaluation similarly to Levenshtein, it first tokenizes the words to capture different aspects of similarity.

Method	Description
`TOKEN_SORT_RATIO`	Converts to lowercase, removes non-alphanumeric characters, sorts tokens alphabetically, and computes a Levenshtein ratio.
`TOKEN_SET_RATIO`	Splits strings into tokens, identifies unique and common tokens, and computes the ratio based on their intersection.

Table 3.6: Description of Utilized RapidFuzz Library Matching Methods

As illustrated in Table 3.6, two methods of this library have been applied. Token_sort_ratio sorts tokens before scoring, and Token_set_ratio compares the intersections between two strings, both of which are well-suited for highly paraphrased facts. The final score produced by the methods used in both Levenshtein and RapidFuzz libraries, are then evaluated using the general threshold logic (Section 3.2) and each fact is labeled accordingly. However, it must be noted that because the string-based approach is fundamentally a lexical metric, it relies on vocabulary intersection. Although being applicable to texts with different structures, especially with token_sort, it cannot capture semantic equivalence if the paraphrased fact uses entirely different vocabulary (synonyms). This limitation motivates the embedding similarity approach in the next section.

3.5 Embedding Similarity Scoring

The string-based approach captures similarity at the character-level or token-level, but it cannot recognize the semantic similarity when different vocabulary or restructured syntax is used (Section 2.4). Embedding similarity scoring addresses this limitation by mapping each sentence to a dense vector where semantically related sentences are placed near each other in vector space.

This thesis utilizes the pre-trained model **"all-MiniLM-L6-v2"** (Reimers and Gurevych, 2019). With a 384-dimensional vector and quick performance, this model provides a balance between semantic representation quality and computational speed, being consistent with the lighter-weight design objectives. The model relies on zero-shot, off-the-shelf semantic matching, without domain specific fine-tuning. This reliance on off-the-shelf models is due to the constraints of the IBFD dataset. Supervised fine-tuning requires explicit, manually annotated pairs, mapping a claim to its evidence. Because the 899 facts have been extracted from the summaries rather than the award documents, plus the small size and extreme imbalanced data, specific fine-tuning of an embedding space is not feasible. Therefore, the approach depends on the model's generalized pre-training.

In this approach, the similarity between a fact and a candidate is computed with cosine similarity. To keep this efficient across roughly 119,000 sentences, the sentence embedding for each document is pre-computed and cached once prior to the scoring step. For each document, the extracted sentences are encoded in batches of 64 and stored as a matrix indexed by sentence position. This pre-computing reduces the computation time to a single forward pass through the encoder plus one sentence matrix multiplication. The cosine similarity scores are then evaluated using the general threshold logic for labeling each fact (Section 3.2). For an in-depth error analysis presented in Chapter 5, the matching sentence is retained.

3.6 Paraphrase Detection Scoring

The embedding similarity scoring approach captures topical and semantic overlap between a fact-sentence pair, but as mentioned at the end of Section 2.4, it cannot reliably distinguish between a genuine paraphrased fact and a fact that shares vocabulary with the source while expressing a different meaning (Table 3.7). The IBFD dataset contains a significant number of paraphrased facts, where the meaning is preserved but the vocabulary or syntactic structure has been altered. Therefore, a paraphrase detection approach is used for targeting this case specifically.

The model used for this scoring approach is **"paraphrase-mpnet-base-v2"** (Transformers, 2024), built on MPNet Song et al. (2020). Unlike `all-MiniLM-L6-v2`, this model is trained on paraphrased pairs and it is used as a bi-encoder, where each fact and candidate is encoded independently and compared by the resulted scores from cosine similarity. Therefore, the model's representation space is directed toward meaning equivalence rather than surface topical similarity. Otherwise, it is utilized identically to the embedding model in Section 3.5. It is an off-the-shelf publicly available bi-encoder model, for the same lighter-weight approach and data-limitation reasons. Moreover, it uses the same pre-compute caching, cosine similarity, and threshold-based labeling logic. Using the same bi-encoder architecture keeps the two approaches comparable by focusing on their main training objectives. The best match candidate is retained for

Fact	Source Sentence	Relationship
The Ministry of Finance confirmed that turbines imported under the PPAs with Electroquil were tax and duty-exempt.	As the Ministry of Finance itself acknowledged, the turbines brought into the country under the power-purchase agreement signed with Electroquil were relieved of taxes.	Meaning preserved
Three key Yukos officers were convicted between July and October 2003.	Between July and October 2003, three key Yukos officers were arrested.	Shared vocabulary, different meaning

Table 3.7: Two paraphrased facts and matched sentence: one where the meaning is preserved despite a different surface form, and another that shares most of its vocabulary while asserting a different meaning. Surface similarity alone does not separate the two cases.

the error analysis in Chapter 5.

3.7 Natural Language Inference Scoring

The string-based comparison, embedding similarity, and paraphrase detection approaches (Sections 3.4 through 3.6), each produce a single score for each fact-candidate pair. The Natural Language Inference (NLI) task has a different structure; instead of simply asking whether two sentences are similar or not, it asks whether one sentence (the premise) logically entails, contradicts, or is neutral with respect to the other sentence (the hypothesis). This makes NLI a suitable approach because it directly aligns with the current task: a candidate sentence from the award document acts as a premise, while the generated atomic fact serves as the hypothesis. The final prediction includes three different scores for each of the mentioned classes, providing insight on whether the source supports or refutes a fact.

The models utilized are **"cross-encoder/nli-deberta-v3-base"** (Sentence Transformers, 2024), and a larger variant **"MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli"** (Laurer et al., 2024). Both are cross-encoder models built based on *DeBERTa-v3* architecture (He et al., 2021a).

The small model is fine-tuned on a combination of NLI datasets including SNLI (Bowman et al., 2015) and MultiNLI. The large model is fine-tuned on MultiNLI plus broader combination of NLI, including: MNLI (Williams et al., 2018), FEVER-NLI, ANLI (Williams et al., 2022), Ling-NLI, and WANLI (Liu et al., 2022). Consistent with choices made in Sections 3.5 and 3.6, the models are used in their publicly available form. For each premise-hypothesis pair, they produce a distribution over the three classes, which are normalized from logits into probabilities using *softmax* before evaluation. Contrary to the bi-encoder approach where each sentence is independently encoded, a cross-encoder processes the premise and hypothesis jointly, activating cross-attention between the tokens of the two sentences Reimers and Gurevych (2019). This joint evaluation architecture is the reason behind the model's ability to examine a logical semantic relationship between the two sentences. However, this also means that each pair (premise, hypothesis) requires an individual forward pass through the model, which increases the computation time. Despite this cost, this approach is included to evaluate whether its deeper logical comparison justifies the processing expense.

3.7.1 Architecture Reasoning

As noted in Section 2.5.2, DeBERTa disentangled attention improves NLI performance over BERT and RoBERTa. Because it captures relative positions explicitly and separately rather than folding them together with the token's content into a single representation, it is in principle better suited for the present task. Considering that case law sentences are long and structurally complex, this approach helps capturing the long-ranged relationships such as negations (common in legal texts), where the decision between Verified and Contradicted lies on structure rather than vocabulary.

Hence, DeBERTa uses a disentangled attention mechanism and for every given token i and j within the concatenated fact-sentence pair, the attention weight $A_{i,j}$ is computed using independent content vectors (H^c) and relative position vectors (P^r):

$$A_{i,j} = \underbrace{H_i^c H_j^{c\top}}_{\text{content-to-content}} + \underbrace{H_i^c P_{i,j}^{r\top}}_{\text{content-to-position}} + \underbrace{P_{j,i}^r H_j^{c\top}}_{\text{position-to-content}}$$

3.7.2 Classification and Scoring

First, the DeBERTa cross-encoder models compare each fact against the sentences in its candidate pool. Once evaluated, the models' three output probabilities are turned into a final binary classification label for each fact ($\hat{Y}_{fact}$). Two distinct aggregation strategies are implemented to interpret the NLI output.

1. **Early-Exit Thresholding:** This strategy assumes a fact is Contradicted unless a candidate supports it. For each fact–candidate pair the entailment probability is the focus; if any candidate's entailment score reaches the threshold θ, the fact is labeled Verified and the remaining candidates are skipped (early exit).

$$\hat{Y}_{fact} = \begin{cases} \text{Verified}, & \text{if } \exists s \in \mathcal{S}_{cand} : P(\text{E} \mid s, F_{target}) \geq \theta \\ \text{Contradicted}, & \text{otherwise} \end{cases}$$

   where $\mathcal{S}_{cand}$ is the candidate pool, E the entailment class, F_{target} the fact, s a candidate, and θ the threshold.

2. **Asymmetric Thresholding:** This strategy instead assumes a fact is Verified unless strongly contradicted. It takes the maximum contradiction and maximum entailment scores over the whole candidate pool, and labels a fact Contradicted only when the contradiction score reaches θ *and* the entailment score stays below it, so facts with contested candidates are not prematurely discarded.

$$\hat{Y}_{fact} = \begin{cases} \text{Contradicted}, & \text{if } \max_{s \in \mathcal{S}_{cand}} P(\text{C} \mid s, F_{target}) \geq \theta \\ & \quad \wedge \max_{s \in \mathcal{S}_{cand}} P(\text{E} \mid s, F_{target}) < \theta \\ \text{Verified}, & \text{otherwise} \end{cases}$$

   where C is the contradiction class and other symbols are as above.

The two strategies examine the trade-off between full semantic comparison and computational cost: early exit is cheaper, while asymmetric thresholding inspects the full candidate pool for each fact.

3.8 IBFD LLM-Based System

The current lighter-weight approach (Sections 3.2) is evaluated against the existing fact verification system developed by IBFD, which serves as the upper-bound baseline for this thesis. The IBFD system is an LLM-based fact verification configured to process each fact against the award documents. It utilizes GPT-5.4 mini, an LLM optimized for fast reasoning. For every fact, the system retrieves the top candidate chunks of sentences from the source document, using the LLM to compare facts against those chunks. The prompt directs the model to also focus on entity-level hints common in legal texts, such as names, dates, and locations, and to assess whether these hints match between the fact and the retrieved chunks (see Appendix B). According to this comparison, the system outputs three classes, the third of which (Insufficient Evidence) is merged into Contradicted for the task's binary evaluation, and assigns one of these two labels: *Verified* if the hints match, *Contradicted* if they reject one another (Table 3.8). Unlike the approaches developed in this thesis, the IBFD system does not produce an intermediate fact-candidate similarity or entailment score.

Document	**Extracted Fact**	**LLM-based System Verification**	**Human Verification (Gold)**
9REN v. Spain	"The date of the celebration is 29 June 2019."	Verified	Contradicted
Ampal v. Egypt	"Breaches of international investment agreement were found"	Verified	Verified

Table 3.8: Examples of Labeled Fact Instances and Verification Statuses

The IBFD system's predictions are used as fixed results without the need to re-run the model. Therefore, no parameter tuning, prompt engineering, or modification to IBFD's system is performed in this thesis. Moreover, the LLM baseline prompt was developed and iterated using the same summaries and award documents utilized for this task.

3.9 Chunk Construction

As the LLM-based system compared the facts against chunks of sentences in the award documents, a further comparison was performed by moving from sentence-level to chunk-level comparison. For this evaluation, the extracted texts were grouped into multi-sentence chunks to provide the scoring models with broader surrounding context. The boundaries were determined using spaCy, followed by a cleaning pre-processing step to remove common OCR problems such as footnotes and bullet markers, by using regex. Sentences were then merged into chunks based on a strict token budget measured by `all-MiniLM-L6-v2` tokenizer. Each chunk was capped at 150 tokens. Also, to prevent losing information required for verification, consecutive chunks were designed with an overlap of approximately 30 tokens.

3.10 Evaluation

The evaluation metrics for this thesis compare the results of the lighter-weight approach against IBFD's LLM-based system baseline. Each fact is treated as a single unit, with one gold label and one predicted label per fact. The matched candidate selected during the scoring is retained only for diagnostic purposes of the error analysis (Chapter 5), and is not itself evaluated. Throughout this thesis, the evaluation is examined in the binary setting, in which the 20 Insufficient Evidence instances are merged into Contradicted (Section 3.1.2), leaving two classes: Verified and Contradicted. Per-class precision, recall, and F1 score are presented for each scoring approach, alongside macro-average of the two classes. Macro-F1 score is the primary metric of comparison used throughout this thesis, as the dataset is heavily imbalanced (roughly 5:1 Verified to Contradicted). Therefore, a metric that weights both classes equally is necessary to assess the performance of the approaches, particularly for the Contradicted class as the smaller of the two.

Each scoring approach produces a continuous score in a specific range, between [0, 100] for RapidFuzz and [0, 1] for all other approaches, so the threshold is treated as a per-approach hyperparameter rather than a fixed value for all approaches (see Appendix C). The dataset provided by IBFD is proposed as a test set and is not split into a separate development set used for threshold tuning. Therefore, the thresholds reported for each approach are the values of the best macro-F1 score on this dataset. Although this lacks out-of-sample validation, it is justified by the exploratory nature of this thesis, as the main objective is to evaluate the best achievable performance of each individual scoring approach.

The lighter-weight approach and the IBFD baseline are both evaluated on the same dataset of 899 facts. The four approaches are compared independently against the baseline, providing an answer to the three sub-questions (Section 1.2). String-based comparison addresses Sub-question 1, the embedding similarity approach addresses Sub-question 2, and paraphrase detection together with NLI addresses Sub-question 3.

In addition to sentence-level comparison, a chunk-level comparison was performed for all four approaches, where the source text is segmented into multi-sentence chunks following the IBFD system's candidate format, to test whether this affects the performance. The results for both settings are reported and compared against the baseline in Chapter 4.

Chapter 4

Results

4.1 Evaluation Setup

This chapter reports the results of the lighter-weight approaches (Chapter 3) and compares the performance of each approach against the IBFD LLM-based upper-bound baseline. Each approach is evaluated at the threshold yielding its highest macro-F1 score. For the string-based comparison, embedding similarity, and paraphrase detection approaches, results are reported both with and without the NER pre-filter to assess its effect. Because the cross-encoder NLI model requires a separate forward pass per fact-candidate pair, running it on the full dataset is not computationally feasible; therefore, NLI is reported with the NER pre-filter only. For the other approaches, filtering does not lower the macro-F1 score (Sections 4.2-4.4.1), therefore, it is assumed that it is also not detrimental for NLI, though this is not checked directly and is noted as a limitation (Chapter 6).

To address the three research sub-questions (Section 1.2), Section 4.2 evaluates the string-based comparison approach (Sub-question 1), Section 4.3 the embedding similarity approach (Sub-question 2), Section 4.4 paraphrase detection and NLI (Sub-question 3). Section 4.5 examines whether chunk-level fact verification (merging multiple sentences into a chunk) affects the overall performance. Finally, Section 4.6 addresses the main research question by comparing each lighter-weight approach against IBFD's upper-bound baseline. Because the dataset is highly imbalanced (Section 3.10) , the macro-averaged F1 score is used as the primary metric throughout this chapter, with per-class precision and recall also reported.

4.2 String-Based Comparison

This section reports the results of the string-based comparison approach based on edit-distance scoring. Before evaluating the string-based approach, the simplest baseline of comparing similarity is examining exact sub-string match, in which a fact is classified as verified only if its text appears exactly inside the candidates from the source documents. This was applied to the full dataset and showed that only 2 verified facts matched at the sentence-level (and 4 at chunk-level; see Section 3.9) out of 899 facts, with no contradicted matches in either setting. This result confirms what was stated in Section 3.4: the facts are rarely plain verbatim restatements of source sentences, but paraphrased versions in which the vocabulary, syntactic structures, or both have been

altered. This motivates moving towards approximate string matching approaches, using Levenshtein and RapidFuzz.

Class	Precision	Recall	F1-Score	Support
Verified	0.830	1.000	0.908	747
Contradicted	0.000	0.000	0.000	152
Macro Avg	0.415	0.500	**0.454**	899

Table 4.1: Performance when predicting all facts as the majority class (Verified).

Beyond the exact-match check, it is necessary to show the baseline based on the dataset's class imbalance. Table 4.1 shows the performance of a naive classifier when every fact is predicted as Verified. The Verified-class receives an F1-score of 0.91 while the macro-F1 score is 0.45.

			Verified			Contradicted			
Method	NER Filter	Threshold	P	R	F1	P	R	F1	Macro-F1
Levenshtein (`ratio`)	Yes	0.5	0.847	0.907	0.876	0.303	0.197	0.239	**0.557**
	No	0.5	0.834	0.942	0.884	0.218	0.078	0.115	0.500
RapidFuzz (`token_sort_ratio`)	Yes	55	0.856	0.785	0.819	0.252	0.355	0.295	**0.557**
	No	55	0.845	0.840	0.842	0.237	0.243	0.240	0.541

Table 4.2: Precision (P), Recall (R), and F1-score for the two string-based scoring libraries at their macro-F1 optimal thresholds, comparing performance with and without the NER pre-filter.

Table 4.2 reports the per-class and macro-F1 scores for both Levenshtein and RapidFuzz, showing the results for the threshold providing the highest macro-F1 score. With the NER pre-filter, both perform almost identically, each reaching a maximum macro-F1 score of 0.56. The per-class breakdown shows that Levenshtein achieves a higher F1 score on the Verified class (0.88 against RapidFuzz's 0.82), while RapidFuzz achieves a higher F1 score on the Contradicted class (0.30 against Levenshtein's 0.24). Although the Verified and Contradicted class scores differ between them, the overall per-class breakdown shows that both libraries perform better for Verified class and do not perform well for Contradicted class. Adjusting the threshold does not resolve this, because while increasing it recovers more Contradicted class facts, it decreases Verified recall and overall macro-F1 score. This is consistent with the overall design of Levenshtein and RapidFuzz: both examine the similarity mainly on character-level edit distance, and RapidFuzz's token_sort tokenization and normalizing the word order (Section 3.4) manages to slightly shift the balance towards the Contradicted class but without changing overall performance.

The NER pre-filter's effect is mainly on the Contradicted class. The number of correctly identified Contradicted facts increases with the filter, from 12 to 30 cases for Levenshtein and from 37 to 54 for RapidFuzz (out of 152), which is reflected in the Contradicted recall and F1 in Table 4.2. By looking at Levenshtein, it can be seen that the performance on the Verified class barely changes (0.884 to 0.876). Considering macro-F1 and this minimal change, the model's improved ability to capture Contradicted class facts results in a noticeable increase in the overall macro-F1 score, pushing it from 0.50 to 0.56. Although RapidFuzz also improved on the Contradicted class, the

Verified class F1 score decreases from 0.842 to 0.819. Therefore, the increase in macro-F1 score is only from 0.54 to 0.56. In addition, both libraries reach an F1 score of the Verified class above 0.80, but neither manages to raise the F1 score of the Contradicted class above 0.30. The Contradicted class is consistently the harder class for both approaches to capture, and the reasons behind these misclassifications are examined in the error analysis (Chapter 5).

4.3 Embedding Similarity

This section reports the results of the embedding similarity approach, which uses `all-MiniLM-L6-v2` sentence transformer and computes the similarity scores with cosine similarity (Section 3.5).

Table 4.3 reports the results for the macro-F1 optimal threshold, with and without NER pre-filtering.

			Verified			Contradicted			
Method	NER Filter	Threshold	P	R	F1	P	R	F1	Macro-F1
`all-MiniLM-L6-v2`	Yes	0.65	0.854	0.816	0.835	0.259	0.315	0.284	**0.559**
	No	0.65	0.849	0.874	0.861	0.276	0.236	0.255	0.558

Table 4.3: Precision (P), Recall (R), and F1-score for the embedding similarity approach at its macro-F1 optimal threshold, comparing performance with and without the NER pre-filter.

Using embedding similarity scoring, a macro-F1 score of 0.56 is reached which is approximately the same as the string-based approach. Considering these results, dense semantic embeddings provide no improvement over the string-based approach on this dataset. The per-class breakdown shows no consistent advantage over string-based comparison approach. Embedding similarity reaches a Contradicted-class F1 of 0.28, putting it between Levenshtein's 0.24 and RapidFuzz's 0.30, and a Verified-class F1 in a similar range for both. Therefore, it neither improves Contradicted classification over the best string-based configuration, nor changes the macro-F1 score which remains at approximately 0.56. Moreover, like the string-based approach, the NER pre-filter's effect is mainly on the Contradicted class. The number of correctly classified Contradicted facts increases from 36 without the filter to 48 with it (out of 152) which is reflected in recall and F1 in Table 4.3. However, this improvement is offset by a simultaneous drop in the Verified class F1 score (from 0.861 to 0.835). So, this provides almost no increase in macro-F1 score (0.558 to 0.559). Therefore, the filter is retained for both consistency across approaches and for computation time discussed in Section 3.2. Adjusting the threshold does not resolve the low macro-F1 score, because raising it from 0.65 to for example 0.75 does increase Contradicted recall, from 0.31 to 0.60, but at the cost of Verified recall which falls from 0.82 to 0.53.

Overall, on this dataset, embedding similarity does not improve on string-based comparison. In this scoring approach, the Contradicted class F1 (0.28) falls between Levenshtein and RapidFuzz, and its macro-F1 score is approximately identical at 0.56, with no consistent per-class advantage for either class. The difficulty remains in detecting the small edits that distinguish Contradicted facts. The reasons behind embedding scoring failing to capture these cases more effectively are presented in the error analysis

(Chapter 5).

4.4 Paraphrase Detection and Natural Language Inference

This section reports the results of the two more semantically informed approaches – paraphrase detection (Section 4.4.1) and natural language inference (Section 4.4.2)– to address sub-question 3. These two approaches are evaluated as described in Sections 3.6 and 3.7. Paraphrase detection uses the `paraphrase-mpnet-base-v2` bi-encoder, and NLI uses cross-encoder models fine-tuned on large-scale NLI datasets (such as SNLI and MultiNLI). Two NLI models are assessed in order to test whether model training data and size affect the performance on this task: a smaller `cross-encoder/nli-deberta-v3-base` model, and a larger `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli`.

4.4.1 Paraphrase Detection

Table 4.4 reports the results for paraphrase detection at its macro-F1 optimal thresholds.

			Verified			Contradicted			
Method	NER Filter	Threshold	P	R	F1	P	R	F1	Macro-F1
`MPNet`	Yes	0.64	0.854	0.848	0.851	0.280	0.289	0.284	**0.568**
	No	0.64	0.850	0.898	0.873	0.309	0.223	0.259	0.566

Table 4.4: Precision (P), Recall (R), and F1-score for the paraphrase detection scoring approach at its macro-F1 optimal threshold, comparing performance with and without the NER pre-filter.

Using the MPNet paraphrase detection model, a macro-F1 score of 0.57 is reached (with NER filter), with an F1 score of 0.85 for the Verified class and a 0.28 F1 score for the Contradicted class. The macro-F1 score is higher than both the embedding similarity (0.56) and the string-based comparison (0.56) results, but the difference is within roughly 0.01 macro-F1 point. Similar to the previous approaches, the NER pre-filter's effect is mainly on the Contradicted class, where correctly classified Contradicted facts rise from 34 without the filter to 44 with it (out of 152). This is reflected in the Contradicted recall and F1 in Table 4.4. Similar to the embedding approach, this gain is almost entirely canceled out by a decrease in the Verified class F1 score (from 0.873 to 0.851). This provides almost no increase in macro-F1 score (0.566 to 0.568). Therefore, a bi-encoder paraphrase-tuned model does not provide substantive improvement over a general-purpose bi-encoder embedding on this dataset, even though the dataset contains a large number of paraphrased facts.

Adjusting the threshold does not resolve this problem, and raising it from 0.64 to, for example, 0.75 increases the Contradicted recall from 0.29 to 0.72, but decreases the Verified recall from 0.85 to 0.46, lowering macro-F1 score from 0.57 to 0.46. The reason the paraphrase detection approach fails to outperform a general embedding model and character-level comparison in this dataset is examined in Chapter 6.

4.4.2 Natural Language Inference

Table 4.5 reports the results for the two NLI models at their macro-F1 optimal thresholds. Two strategies were evaluated (Section 3.7): the early-exit strategy and an alternative asymmetric strategy. The early-exit strategy gave the higher macro-F1, and all NLI results reported in this section are based on its results.

			Verified			Contradicted			
Method	NER Filter	Threshold	P	R	F1	P	R	F1	Macro-F1
`nli-deberta`	Yes	0.3	0.845	0.845	0.845	0.237	0.237	0.237	0.541
`MoritzLaurer`	Yes	0.2	0.865	0.729	0.791	0.249	**0.441**	**0.318**	**0.555**

Table 4.5: Precision (P), Recall (R), and F1-score for the NLI scoring approach at macro-F1 optimal thresholds, comparing the small and large models with the NER pre-filter.

As noted in Section 4.1, NLI results are reported with the NER pre-filter only, considering the per pair separate forward pass cost through the cross-encoder. The NLI scoring approach reaches a macro-F1 of 0.54 with the smaller `cross-encoder/nli-deberta-v3-base` and roughly 0.56 with the larger `MoritzLaurer/DeBERTa-v3-large-mnli-fever-anli-ling-wanli` model. Both are in the same range as the previously mentioned approaches, so at the sentence-level NLI does not improve over other approaches, which is contrary to what the legal-NLI literature would suggest (Section 2.5.2). The per-class results show that the models reach almost similar macro-F1 scores but through different per-class trade-offs. The smaller model has an F1 score of 0.84 for the Verified class and 0.24 for the Contradicted class, which for this threshold are balanced and near identical for each class. On the other hand, the larger model results show that it performs better for the Contradicted class, but at the cost of lower Verified F1. It achieves the highest Contradicted class F1 of any sentence-level approach examined in this thesis (0.32), driven by higher recall on the Contradicted class (0.44), but at the cost of lower recall (0.73) and F1 score (0.79) on the Verified class.

The higher Contradicted recall comes with low Contradicted precision (0.25). The model labels a larger share of facts Contradicted, so it captures more correctly classified Contradicted facts by simply predicting the class more often, rather than reliably distinguishing Contradicted from Verified facts. This is the first approach to increase the Contradicted class recall and F1 score, which suggests the NLI entailment, contradiction, and neutral signals do add something the other approaches lack, even though it does not raise the overall macro-F1 score. Raising the threshold has a similar effect as other approaches. At a higher threshold of, for example, 0.8, the smaller model reaches Contradicted recall 0.32 (macro-F1 0.52) and the larger model reaches 0.76 (macro-F1 0.45). The larger model's higher Contradicted recall confirms the two models distribute their predictions differently, but both suffer macro-F1 score decrease in higher thresholds. One reason for the low Contradicted recall is that the information needed to contradict a fact is often spread across more than one sentence; a sentence-level evaluation fails at capturing this. Section 4.5 examines this directly by replacing the sentence-level candidates with chunk-level candidates.

4.5 Sentence-Level vs. Chunk-Level Evaluation

The results reported so far all used sentence-level candidates for comparing the facts against. As noted in Section 3.9, an additional evaluation was conducted on the chunk-level basis, where each chunk consists of several consecutive sentences from the source document. This section examines how that change in granularity affects each approach. This chunk-level setting was motivated by two reasons. First, the facts were generated from case law summaries, where a single summary sentence may compress information spread across several sentences of the source award document. Therefore, the evidence needed to classify a fact may be distributed across more than one sentence, where a single-sentence candidate cannot capture due to incomplete evidence. Second, chunk-level candidates match the input granularity of the IBFD baseline (Section 3.8), allowing a fairer comparison between the two systems.

As a first comparison between sentence-level and chunk-level, exact sub-string matching rises from 2 matches at sentence-level to 4 at chunk-level. This is still negligible against 899 facts, but consistent with chunks capturing slightly more complete text containing the information relevant to the fact.

Method	Sentence Macro-F1	Chunk Macro-F1	Gain/Loss
Levenshtein (`seq_ratio`)	0.557	0.530	-0.027
Embedding (`all-MiniLM-L6-v2`)	0.559	0.559	0.00
Paraphrase (`MPNet`)	0.568	0.556	-0.012
`Small NLI` (`nli-deberta-v3`)	0.541	0.531	-0.010
`Large NLI` (`DeBERTa-v3-large`)	0.555	0.641	**+0.086**

Table 4.6: Macro-F1 for each scoring approach at sentence-level and chunk-level granularity. The gain/loss column illustrates that expanded context slightly degrades or stagnates surface-level and small models, while boosting the Large NLI performance.

Table 4.6 divides the macro-F1 score of the approaches into two groups. For the string-based comparison, embedding similarity, and paraphrase detection approaches, moving to chunk level changes macro-F1 by at most about 0.03 F1 score points in either direction: Levenshtein drops slightly (0.557 to 0.530), embedding is unchanged (0.559), and paraphrase drops marginally (0.568 to 0.556). For these similarity comparison approaches, the larger candidate neither helps nor harms performance to any meaningful degree. The NLI models behave differently. The larger MoritzLaurer model improves noticeably, from 0.555 at sentence level to 0.641 at chunk level; a gain of roughly 0.09 macro-F1 score points, the largest granularity effect in this chapter and the only clear improvement from chunking. The smaller nli-deberta model does not follow the same path: it decreases by 0.01 macro-F1 score point (0.541 to 0.531), essentially flat. So, the benefit of added context is specific to the larger model, not a general property of the cross-encoder architecture.

Table 4.7 shows where the larger model's gain comes from. Both classes improve at chunk level: the Verified class F1 increases from 0.79 to 0.87, and the Contradicted class F1 from 0.32 to 0.42; the highest contradicted class result of any configuration in this study. The improvement is driven by higher precision and recall on both classes rather than a shift in the model's bias toward one class, so the larger candidate allows the model to make better-supported judgments overall, rather than simply relabeling. Why the larger context helps this model in particular is considered in the discussion

	Verified			Contradicted			
`Large NLI` Context	P	R	F1	P	R	F1	Macro-F1
Sentence-level	0.865	0.729	0.791	0.249	0.440	0.318	0.555
Chunk-level	0.885	0.847	**0.866**	0.380	0.461	**0.417**	**0.641**

Table 4.7: Per-class precision (P), recall (R), and F1-score comparison for the `Large NLI` model. Expanding the context window from sentences to chunks improved both precision and recall across the Contradicted class, driving the overall Macro-F1 surge.

(Chapter 6).

4.6 Comparison Against IBFD Baseline

This section addresses the main research question: how do alternative lighter-weight approaches compare to IBFD LLM-based system performance? Here each approach is represented by its highest macro-F1 configuration from the preceding sections, and the IBFD system serves as the upper-bound baseline. The IBFD system's predictions are evaluated against the gold labels using the same per-class metrics applied to the lighter-weight approach, on the binary evaluation set (Section 3.1.3) presented in Table 4.8.

Class	Precision	Recall	F1-Score	Support
Verified	0.940	0.857	0.896	747
Contradicted	0.509	0.730	0.600	152
Accuracy			0.835	899
Macro Avg	0.724	0.794	0.748	899
Weighted Avg	0.867	0.835	0.846	899

Table 4.8: IBFD LLM-based system performance on the binary evaluation set

The baseline reaches a macro-F1 score of 0.75. Its performance is different across classes: an F1 score of 0.90 on the Verified class but 0.60 on the Contradicted class. This is the same asymmetry seen throughout the lighter-weight results, where the Contradicted class is harder to capture, and it indicates that difficulty is a property of the task and dataset rather than of any single system, since it persists even for the LLM-based baseline.

The macro-F1 score of 0.75 is the reference against which the lighter-weight approaches are compared. Table 4.9 shows two levels of performance. Five of the six approaches – both string-based comparison methods, embedding similarity, paraphrase detection, and the smaller NLI model – cluster closely between 0.54 and 0.57 macro-F1 score, which is below the baseline performance on macro-F1 score (0.75), but above the majority-class floor of 0.45. The larger NLI model at chunk-level stands apart at 0.64, the closest any configuration comes to the LLM-based system. The answer to the main research question has two parts. First, no lighter-weight configuration matches the IBFD baseline at 0.75 on this dataset, but the best configuration is the chunk-level scoring with the large NLI model at 0.64, without the API cost of an LLM-based system. Second, the cheapest approaches – string-based comparison, embedding similarity,

Method	Best Configuration	Macro-F1
Majority-class	Sentence/Chunk, $\theta = 0.0$	0.454
Levenshtein (`ratio`)	Sentence, $\theta = 0.5$	0.557
RapidFuzz (`token_sort_ratio`)	Sentence, $\theta = 55$	0.557
Embedding (`all-MiniLM-L6-v2`)	Sentence, $\theta = 0.65$	0.559
Paraphrase (`MPNet`)	Sentence, $\theta = 0.64$	0.568
`Small NLI` (`nli-deberta-v3`)	Sentence, Early-Exit, $\theta = 0.3$	0.541
`Large NLI` (`DeBERTa-v3-large`)	Chunk, Early-Exit, $\theta = 0.95$	**0.641**
IBFD baseline	Chunk	0.750

Table 4.9: Summary of the macro-F1 optimal configuration for each scoring approach compared against the IBFD baseline. The Large NLI model operating at the chunk-level achieves the highest overall performance.

and paraphrase detection – manage to obtain a macro-F1 score of around 0.56 while avoiding both the API cost of an LLM and the per-pair cost of a cross-encoder.

Overall, the chapter's central finding is that a lighter-weight approach can get close but not match an LLM-based fact verification system on the evaluated case law documents, with chunk-level NLI the closest at 0.64. The error patterns underlying this performance are examined in Chapter 5.

Chapter 5

Error Analysis

The results reported in Chapter 4 show that the evaluated approaches get close to the IBFD baseline macro-F1 score (0.75), and chunk-level NLI reaches 0.64, although a gap remains. Across every approach, the Contradicted class consistently scores lower F1 than the Verified class. This chapter examines why these approaches misclassify facts by inspecting the misclassifications and identifying systematic patterns. It argues that these errors are largely caused by the structure of the dataset, having Contradicted class facts that are minimally edited and paraphrased from source documents while remaining lexically close to them. This reflects how errors realistically arise in summaries, as small alterations rather than complete fabrication. The false positive cases (Contradicted facts misclassified as Verified) are the focus of this chapter because the Contradicted class is both the hardest to capture and more practically important, since detecting incorrect facts is a primary purpose of a fact verification system in the context of IBFD. For the string-based comparison, embedding similarity, and paraphrase detection approaches, the errors are grouped into five error categories derived from manual inspection of the misclassifications:

- **Numeric/Temporal:** Facts closely similar in wording to a source sentence, but differing in a numeric or temporal value. This includes cases where a number, quantity, or date value differs outright, and cases where a date expression partially overlaps – one component (typically the year) matches and inflates the similarity score while another (the month or day) differs – leaving the fact unverified.
- **Semantic Mismatch:** Facts in which the surface structure is broadly similar to the source, but a key content word has been replaced with a word carrying an opposite or different meaning.
- **Negation:** Facts that share the main vocabulary with the source sentence but differ in negation scope or polarity and indicate reverse meaning.
- **Abbreviation:** Facts in which a sequence of words in the source sentence is presented as an abbreviated form.
- **Topic Overlap:** Facts and sentences share similar words, like party names or headings, but assert nothing that verifies it.

These categories are illustrative; therefore, not all appear for every approach, and a number of false positives are caused by sentence-segmentation problems. The NLI

approach is analyzed separately (Section 5.4) because it produces an entailment judgment and errors mostly concern the models' scoring judgment rather than what differs between the texts. These categorizations were made by the author alone.

The error analysis is performed at the level of the matched candidate and for each false positive, the specific candidate responsible for the decision (incorrect Verified) is retained, making it possible to examine the reason for the score each pair received. This traces the model's decision and not how gold labels were annotated. Errors are grouped based on scoring approaches rather than individual library or model, as the patterns reflect the characteristic of the underlying mechanism itself. Sections 5.1 through 5.4 examine string-based comparison, embedding similarity, paraphrase detection, and NLI. Section 5.5 evaluates the findings together, indicating that the limiting factor is not the choice of scoring approach but the properties of the dataset: the class imbalance (roughly 5:1 for Verified), the minimally edited paraphrased Contradicted facts which keeps them lexically close to the source, and the existence of lexically similar sentences across each document.

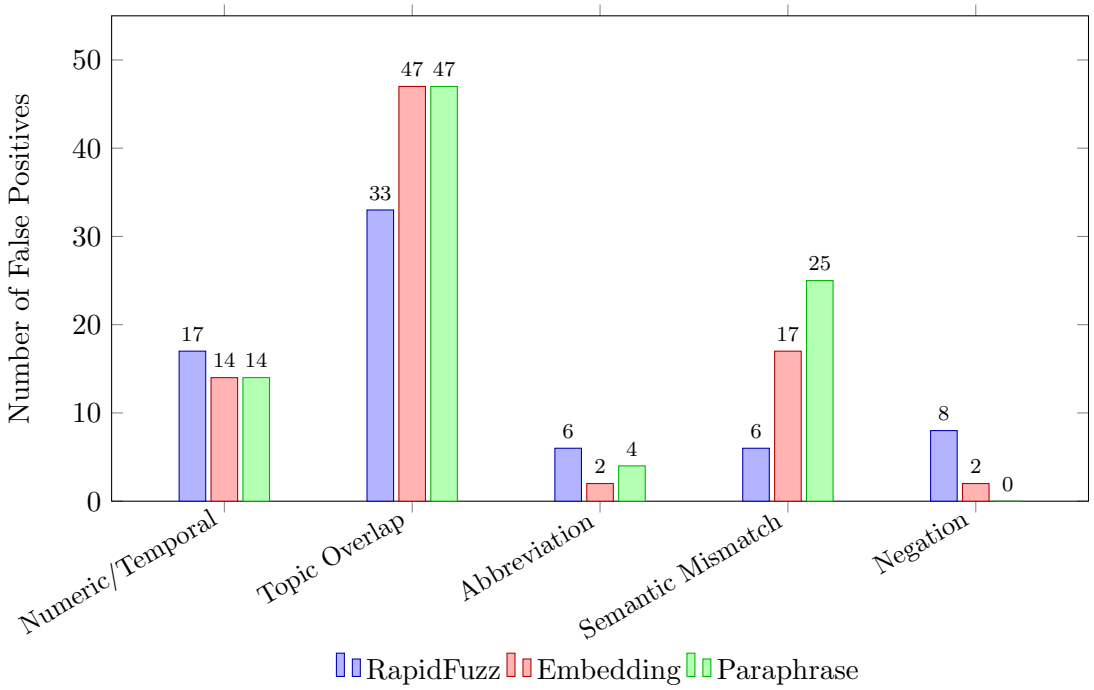


Figure 5.1: Distribution of false positive error categories across string-based, embedding, and paraphrase detection approaches.

Figure 5.1 shows the distribution of all false positive cases across the noted five error categories for string-based comparison, embedding similarity, and paraphrase detection approaches. Errors related to segmentation failure are treated separately as a shared pre-processing reason (Section 5.5.1), therefore the counts are fewer than each approach's total false positives. NLI errors are also treated separately (Section 5.4) as they are not on surface-level and the errors are categorised differently.

5.1 String-Based Comparison

This section examines the misclassifications produced by the two Python libraries used for the string-based scoring approach, Levenshtein and RapidFuzz (Section 3.4). Both libraries were independently inspected, and the false positive cases were extracted and grouped according to the five defined categories. Because both libraries share the same character-level mechanism, they produce almost similar error patterns. Therefore, the examples presented in this section are captured from the RapidFuzz `token_sort_ratio` method for threshold 55, which is the best string-based performing configuration for the Contradicted class in Chapter 4. Of the five categories (Figure 5.1), Topic overlap errors are the most frequent. The confusion matrix for the macro-F1 optimal threshold is illustrated in Figure 5.2

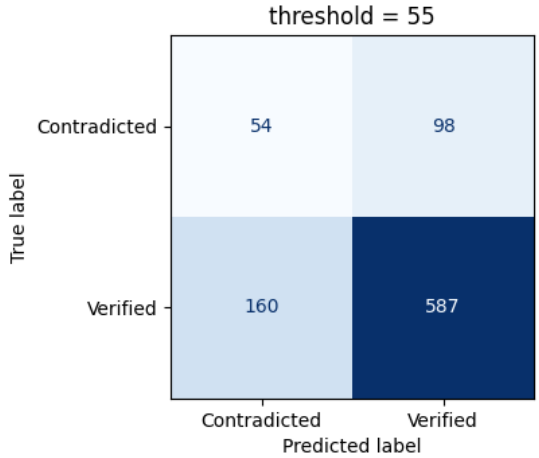


Figure 5.2: RapidFuzz Threshold 55 Confusion Matrix with 98 False Positives

- **Numeric/Temporal:** Because the string-based approach calculates the similarity based on character overlap, it is vulnerable to numerical changes. If a fact and a sentence are similar except for one or two digits, their similarity score would be high, resulting in a false positive.

  Fact: *CEMSA started exporting cigarettes in 1980.*
  Sentence: *The claimant's firm, CEMSA, first began exporting cigarettes in 1990.*
  Similarity score: 64.3
  Analysis: 1980 vs 1990

  Fact: *Law 24 amended the Hydrocarbons Law*
  Sentence: *Law No.2006-42 Amending the Hydrocarbon*
  Similarity score: 66.9
  Analysis: 24 vs 42

  Fact: *The tax authorities initiated investigations in March 2002.*
  Sentence: *The tax authorities initiated their investigation of Taki spravy on 14 February 2002, which*
  Similarity score: 69.3
  Analysis: March vs February

- **Negation:** For the same character overlap reason, when a fact and a sentence share similar words but are different in the presence or scope of negation, their similarity score would be high, resulting in a false positive.

  Fact: *Law 42 with a 99% tax rate did not substantially deprive Burlington of the value of its shares.*
  Sentence: *it substantially reduced on account of Burlington's behavior leading to its loss.*
  Similarity score: 55.9
  Analysis:— did not substantially deprive vs substantially reduced

  Fact: *The Tribunal ruled that the damages for the revocation of the EMG licence must be calculated beyond 2025.*
  Sentence: *revocation of EMG's tax license will be calculated up to 2025 and not beyond.*
  Similarity score: 67
  Analysis: beyond vs not beyond

- **Semantic mismatch:** Facts and sentences are lexically close but differ in a content word that changes the meaning.

  Fact: *Three key Yukos officers were convicted between July and October 2003.*
  Sentence: *Between July and October 2003, three key Yukos officers were arrested. In July 2003,*
  Similarity score: 64.9
  Analysis: convicted vs arrested

  Fact: *TS printed a small number of election campaign materials*
  Sentence: *Taki spravy printed a large amount of campaign materials for Yulia Tymoshenko*
  Similarity score: 57.6
  Analysis: small number vs large amount

- **Abbreviation:** Where an acronym in the fact has no expanded or relevant counterpart in the sentence.

  Fact: *GEM paid all VAT amounts assessed under the first tax assessment.*
  Sentence: *and GEM subsequently paid all the amounts assessed under the first Tax Assessment*
  Similarity score: 63.5
  Analysis: VAT is an abbreviation which itself and its expanded version is not mentioned in the sentence.

- **Topic Overlap:** When the sentence shares some words with the fact – a party name, a block number, or a stock phrase – but otherwise makes a different statement. Because Token_sort sorts the tokens and compares them, these instances receive a similarity score higher than the best-performing threshold.

Fact: *Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.*
Sentence: *C. Perenco's interests in the Participation Contracts for Blocks 7 and 21*
Similarity score: 57
Analysis: The match is a result of having similar words between the fact and the sentence, while the sentence is essentially a heading and does not mention anything about oil prices dropping.

Fact: *TS was the target of a long-running campaign of oppression by international agencies*
Sentence: *This proposition is that Taki spravy was the target of a long-running campaign of oppression*
Similarity score: 70.5
Analysis: There are many similar tokens present between the fact and the sentence, but there is no mention of the "international agencies".

5.2 Embedding Similarity

This section examines the misclassifications produced by the embedding similarity approach (Section 3.5). The errors are examined according to the threshold 0.65 (with NER pre-filter) which is the best-performing embedding similarity configuration for the Contradicted class in Chapter 4. Of the five error categories (Figure 5.1), topic overlap error is the most frequent. The confusion matrix for the macro-F1 optimal threshold is illustrated in Figure 5.3

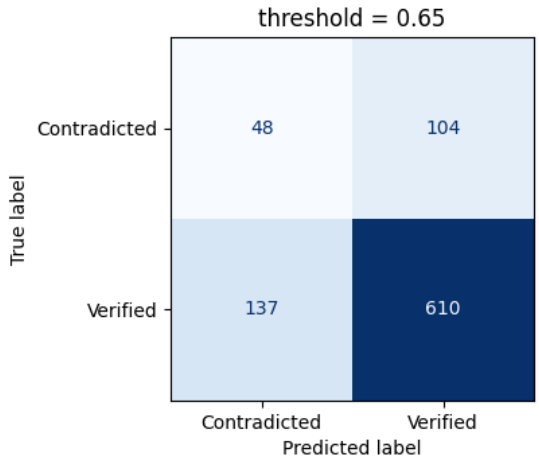


Figure 5.3: Embedding Threshold 0.65 Confusion Matrix with 104 False Positives

- **Numeric/Temporal:** Because a changed number barely shifts the embedding, when fact and sentence express the same proposition but differ in a numeric value, they receive a high similarity score from cosine similarity.

  Fact: *CEMSA started exporting cigarettes in 1980*
  Sentence: *The claimant's firm, CEMSA, first began exporting cigarettes*

in 1990
Similarity score: 0.92
Analysis: 1980 vs 1990

Fact: *Law 24 amended the Hydrocarbons Law*
Sentence: *E. Origins of the dispute: Events relating to Law 42-2006 Amending the Hydrocarbon*
Similarity score: 0.69
Analysis: 24 vs 42

Fact: *The event occurred in May 1998.*
Sentence: *the month of February 1998.*
Similarity score: 0.71
Analysis: May vs February

- **Negation:** Fact and sentence assert similar vocabularies with opposite polarity. A statement and its negation are close in the embedding space, so cosine similarity gives a high similarity score and fails to separate them.

  Fact: *Taxes must be discriminatory.*
  Sentence: *not be discriminatory and they may not be confiscatory.633 Confiscatory taxation*
  Similarity score: 0.69
  Analysis: must be discriminatory vs not be discriminatory

- **Semantic mismatch:** Fact and sentence are close in meaning but differ in a content word – an antonym or point to a wrong referent – that alters the fact. Because antonyms and near-synonyms are close in embedding space, these false positive cases occur.

  Fact: *Three key Yukos officers were convicted between July and October 2003*
  Sentence: *Between July and October 2003, three key Yukos officers were arrested. In July 2003,*
  Similarity score: 0.90
  Analysis: arrested vs convicted

  Fact: *The Czech Republic offered an extended depreciation period for certain assets.*
  Sentence: *exemption and shortened depreciation period. The Czhech Republic*
  Similarity score: 0.83
  Analysis: extended vs shortened

  Fact: *The Mexican President revised the IEPS law effective 1 January 1992.*
  Sentence: *12. Pending final resolution of the Amparo, the Mexican Congress amended the IEPS"*
  Similarity score: 0.74
  Analysis: President vs Congress

- **Topic overlap:** The topical overlap between facts and the sentences occurs, when the matched sentence and the fact are talking about the same subject but assert something different or unrelated. This often occurs for sentences like section headings.

  Fact: *Perenco Ecuador Limited (Perenco) is a company incorporated in Iran*
  Sentence: *PERENCO ECUADOR LIMITED*
  Similarity score: 0.79
  Analysis: The embedding scores the fact and the company's name as highly similar because they share the vocabularies, although nothing is mentioned about where it is incorporated.

  Fact: *The nature of the solar levy requires legal characterization.*
  Sentence: *(ii) Introduction of the Solar Levy*
  Similarity score: 0.79
  Analysis: The embedding scores the Solar Levy very high for similarity, but the matched sentence does not provide related information to verify the fact.

5.3 Paraphrase detection

This section examines the false positive cases produced by the paraphrase detection approach (Section 3.6). For this approach, topic overlap error is the most frequent between the five error categories (Figure 5.1). Figure 5.4 examines the confusion matrix at the threshold 0.64 which is the macro-F1 optimal threshold in Chapter 4.

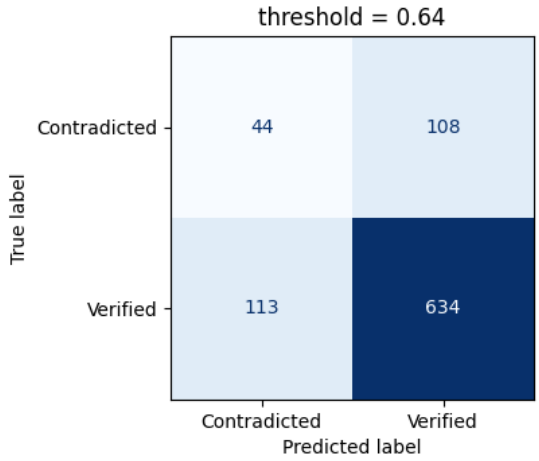


Figure 5.4: Paraphrase Detection Threshold 0.64 Confusion Matrix with 108 False Positives

- **Numeric/Temporal:** Similar to embeddings, when a fact and sentence express the same proposition but differ in a numeric value, the model would consider them similar.

Fact: *CEMSA started exporting cigarettes in 1980.*
Sentence: *The Claimant's firm, CEMSA, first began exporting cigarettes in 1990.*
Similarity score: 0.78
Analysis: 1980 vs 1990

Fact: *INECEL ceased to exist in 1996.*
Sentence: *However, from September 1996 until 31 March 1999, INECEL*
Similarity score: 0.68
Analysis: 1996 vs 1999

Fact: *The event occurred in May 1998.*
Sentence: *the month of February 1998.*
Similarity score: 0.68
Analysis: May vs February

- **Semantic mismatch:** Fact and sentence are close in meaning but differ in a content word, an antonym or a wrong referent, that reverses or alters the fact. The paraphrase model surfaces a notably wide range of antonymic pairs.

  Fact: *Three key Yukos officers were convicted between July and October 2003*
  Sentence: *Between July and October 2003, three key Yukos officers were arrested. In July 2003,*
  Similarity score: 0.89
  Analysis: arrested vs convicted

  Fact: *The participation contracts included a tax stability clause.*
  Sentence: *79. The Participation Contracts also contained a tax modification clause, which required the*
  Similarity score: 0.78
  Analysis: stability vs modification

- **Topic overlap:** The topical overlap between facts and the sentences occurs, when the matched sentence and the fact are talking about the same subject but assert a different or unrelated topic. This often occurs for sentences like section headings.

  Fact: *Perenco Ecuador Limited (Perenco) was engaged in tobacco exploration and exploitation in Ecuador.*
  Sentence: *PERENCO ECUADOR LIMITED*
  Similarity score: 0.85
  Analysis: The matched sentence is only the company's name which results in a high cosine similarity score for the pair, but it asserts nothing about tobacco and whether it is in fact what Perenco was engaged in.

  Fact: *The nature of the solar levy requires legal characterization.*
  Sentence: *(ii) Introduction of the Solar Levy*

Similarity score: 0.74
Analysis: The shared solar levy leads to close vectors and the model considers it as a smaller, paraphrased version, so it would be scored high by cosine similarity.

Fact: *Lone Star made investments in Korean assets between 2001 and 2012.*
Sentence: *Lone Star Principals Involved in the Claimants' Investments in Korea*
Similarity score: 0.78
Analysis: The fact is paired with a section heading on the same subject, so it contains no dates and does not establish the 2001–2012 span.

5.4 Natural Language Inference

The NLI architecture is to perform a premise-hypothesis evaluation and the current task applies it for each fact and across the full candidate pool. With this approach, every fact is scored against the candidates from its source document, and the early-exit strategy (Section 3.7) selects the candidate that first clears the entailment threshold. Therefore, the "sentence" shown for each example is the first candidate selected for that fact.

The error analysis for the natural language inference (NLI) approach is organized differently from the previous analysis. For the string-based comparison, embedding similarity, and paraphrase detection approaches, the error analysis is on the surface level that results in a high score despite topical differences, so the errors are grouped by the type of reasons causing the error (Numeric/Temporal, semantic mismatch, etc.). NLI, on the other hand, does not provide a score based on surface overlaps between the fact and candidate, and produces an entailment (verified) judgment for each fact-candidate pair, treating the candidate as premise and the fact as hypothesis. For NLI, the model outputs an entailment judgment score; therefore, the question is whether that judgment is warranted by the matched candidate; whether the candidate actually verifies the fact or whether the model returned high entailment for a sentence that says nothing relevant (an unwarranted judgment). The NLI errors are analyzed based on two categories. As the task is binary, both NLI error categories below evaluate false positives (Contradicted facts incorrectly labeled as Verified), and they differ in the reason the model made the error.

The first is high-confidence entailment without relevant support, where the matched premise contains no claim relevant to the fact – typically a document header or citation fragment – yet the model returns a high entailment score. Since the score makes it look like a certain decision, this is a more serious failure and when the same premise recurs for several unrelated facts (Appendix A.3), it shows a systematic problem in the entailment scoring. The second is defensible entailment, where the premise actually supports the fact and the high entailment score is reasonable, but the gold label is Contradicted because the contradicting detail is placed outside the matched sentence. As with other approaches, the analysis focuses on false positives at the macro-F1 optimal threshold. All the entailment scores for each pair have been confirmed in a standalone check. Both NLI models (Section 3.7) are analyzed and the results are reported based on the early-exit scoring approach.

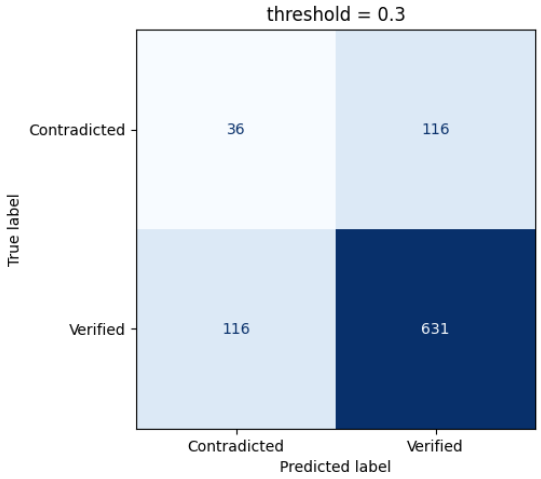


Figure 5.5: NLI Small Model Threshold 0.3 Confusion Matrix with 116 False Positives

5.4.1 NLI Small Model nli-deberta

The smaller model's false positives at the operating threshold (minimum probability to label a fact as Verified) are summarised in Figure 5.5. The two categories are examined in turn below.

- **High entailment without relevant support**

  Fact: *Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.*
  Sentence: *WASHINGTON, D.C.*
  Entailment score: 0.98
  Analysis: The premise is a location line from a document header and says nothing about oil prices, Perenco, or Blocks 7 and 21, yet the model reports 0.98 entailment. The same string is returned for five further unrelated facts (see Appendix A.3), which shows the behaviour is systematic and occurs multiple times rather than a single unusual case.

  Fact: *Yukos emerged as one of the largest oil companies in Russia.*
  Sentence: *Isle of Man (a Dependency of the United Kingdom).*
  Entailment score: 0.99
  Analysis: The premise names an unrelated jurisdiction and contains no claim about Yukos or Russian oil, yet the model returns near-complete entailment.

  Fact: *From then on indicates a point in time after a previously mentioned event*
  Sentence: *Burlington Resources Inc.*
  Entailment score: 0.90
  Analysis: The premise names an unrelated header with a company name and contains no claim about a point in time and an event.

Fact: *CEMSA started exporting cigarettes in 1980.*
Sentence: *1 See the Notice of Intent to Submit a Claim, submitted under NAFTA Article 1119, p. 2.*
Entailment score: 0.89
Analysis: This fact was a false positive under every similarity approach. Here the small model matches it instead to a non-related citation and still returns 0.89 entailment. The fact therefore survives into the NLI stage, but through a different and less excusable failure.

- **Defensible Entailment**

Fact: *The tax authorities initiated investigations in March 2002.*
Sentence: *On 13 March 2002, STS initiated a criminal tax evasion case (Case No 79-00111) against*
Entailment score: 0.99
Analysis: This premise mostly supports the fact – the month, STS as tax authorities, investigation and evasion case No – so the model's high entailment is reasonable. This is instructive against the string-based approach, which matched the same fact to "14 February 2002" (a Numeric/Temporal error). The NLI model retrieves a matched sentence which genuinely entails the fact, however the correct match lies later in the document: "In February 2002, shortly after TS delivered hundreds of thousands of copies of campaign documents, the tax authorities initiated investigations into TS's tax affairs." showing the difference between March and February. This shows two points: there are several closely related sentences in the document which can result in a high entailment but were not the main source, and some of the facts themselves are too general (not mentioning the investigation into what) which can trigger high entailment in candidates other than the source.

Fact: *The event occurred in May 1998.*
Sentence: *1998/05/22*
Entailment score: 0.98
Analysis: The model correctly recognizes that 1998/05/22 falls in May 1998 and entails the fact. The paraphrase detection and embedding, by contrast, matched this fact to "February 1998" (a Numeric/Temporal error). The entailment model handles the date where surface similarity could not, illustrating what NLI adds over the earlier methods. However, because the fact is too general (not specifying the event), it can trigger other candidates with a high entailment.

Fact: *INECEL ceased to exist in 1996.*
Sentence: *October 1996, providing inter alia that INECEL be liquidated within a certain time*
Entailment score: 0.99
Analysis: A liquidation ordered in 1996 reasonably entails that the entity ceased to exist around then, so the model's confidence is defensible. However, the next sentence in the document mentions a certain time "period", indicating a duration and not a certain year; so not exactly 1996.

5.4.2 NLI Large Model MoritzLaurer

Figure 5.6 shows the larger model's false positives. It produces fewer cases than the smaller model (85 against 116), and the high-confidence no-support failures are noticeably less frequent.

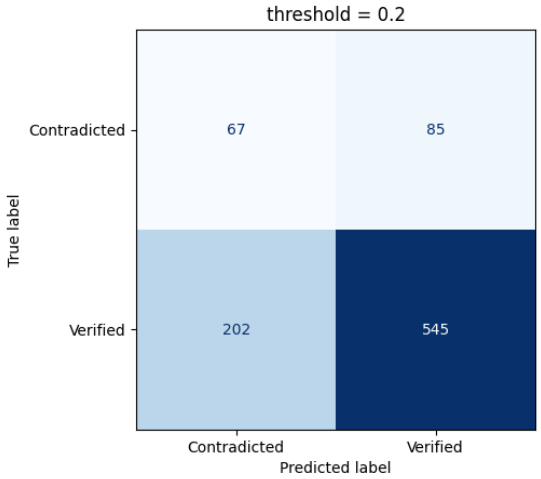


Figure 5.6: NLI Large Model Threshold 0.2 Confusion Matrix with 85 False Positives

- **High entailment without relevant support**

  Fact: *The alternative 80% tax rate law enacted by Ecuador in 2007 avoids the application of Law 42 with a rate of 99%.*
  Sentence: *256. Second, the Blocks were under force majeure status. A declaration of force*
  Entailment score: 0.76
  Analysis: This pair is largely unrelated. The premise concerns force majeure over the Blocks and does not address the alternative 80% tax law or Law 42's 99% rate, yet the model returns 0.76 entailment. This is a failure in capturing the correct candidate, but less severe than the small model's because the sentence is at least from the same dispute rather than a page header.

  Fact: *According to Ecuador, the price increase helped the negotiation of PSCs.*
  Sentence: *256. Second, the Blocks were under force majeure status. A declaration of force*
  Entailment score: 0.54
  Analysis: The same sentence is returned for another unrelated fact about PSC negotiations, at a lower 0.54. That one premise serves as the match for two different facts is the only trace here of the systematic boilerplate behaviour that pervaded the small model.

- **Defensible Entailment**

  Fact: *Yukos emerged as one of the largest oil companies in Russia*
  Sentence: *Yukos became the first Russian company to be ranked among*

the top ten largest oil and
Entailment score: 0.96
Analysis: The premise supports the fact almost directly (Yukos ranked among the largest oil companies), so the entailment is reasonable. The next sentence mentions "[and] gas companies" which might be the reason that the gold label was Contradicted. The small model matched this fact to "Isle of Man" at 0.99 — a pure match failure the large model does not repeat.

Fact: *The tax authorities initiated investigations in March 2002*
Sentence: *On 13 March 2002, STS initiated a criminal tax evasion case (Case No 79-00111) against*
Entailment score: 0.96
Analysis: The model retrieves the correct March 2002 sentence and entails it. The string-based approach had matched this fact to "14 February 2002" (a Numeric/Temporal error). Similar to the small NLI model, both date retrievals are correct, with a more sensible level of confidence by the larger model.

Fact: *The proposal was quickly enforced.*
Sentence: *62. The proposal was accepted and immediately executed. On 2 March 2016, the Minsk*
Entailment score: 0.99
Analysis: "Quickly enforced" and "accepted and immediately executed" are close in meaning, so the near-maximal entailment is reasonable. This is the kind of reasonable near-paraphrase judgment that dominates the large model's false positives: the label is Contradicted based on another sentence, not on a model error.

5.5 Systematic Errors Across Approaches

The preceding sections analyzed each approach's errors separately. One unified pattern can be seen across all approaches, and that is sentence segmentation, where the candidate is truncated before the information that would verify or contradict the fact appears. This truncation is also the mechanism behind the chunk-level improvement (Section 4.5) for the large NLI model, where expanding the candidates from a sentence to a multi-sentence chunk provides more complete information for the NLI model to evaluate. Also, similar false positive cases can be seen across the approaches, showing that none of the approaches managed to label them correctly as Contradicted. Both of these points indicate that the dataset is partly a limiting factor (partly capability gap), as it contains genuinely hard cases (minimally edited, lexically close Contradicted facts) and the sentence-level comparison sometimes lacks sufficient evidence for correct classification.

5.5.1 Sentence Segmentation

By looking at the false positive cases for all the lighter-weight approaches, one common cause of misclassifications comes from how the OCR and sentence split (Section 3.3) provided the sentences extracted from the award document PDFs. Most of the sentences

are not complete, therefore, the full information mentioned in the sentences is not getting captured by the approaches, leading to misclassifications.

- **String-based Comparison**

  Fact: *TS was the target of a long-running campaign of oppression by international agencies*
  Sentence: *This proposition is that Taki spravy was the target of a long-running campaign of oppression*
  Similarity score: 74.5
  Analysis: Most words are similar between the fact and the sentence, but the sentence ends at "oppression" without showing who was in charge of the oppression. In the document, the next sentence mentions "by State agencies"

  Fact: *The Czech Republic committed to an 8% share of electricity from solar energy by 2010.*
  Sentence: *14. The Czech Republic undertook to achieve an 8% share of production of electricity from*
  Similarity score: 69
  Analysis: The sentence ends at "electricity from" dropping the source and the date. The next sentence in the document shows that the data was correct but the source is renewable sources generally, not specifically solar.

- **Embedding Similarity**

  Fact: *Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.*
  Sentence: *81. Shortly after Perenco purchased its interests in Blocks 7 and 21, international oil prices began*
  Similarity score: 0.72
  Analysis: A near-perfect match cut at "international oil prices began," exactly before stating whether they rose or fell, which in the document the next sentence mentions "rise dramatically".

  Fact: *The Czech Republic committed to an 8% share of electricity from solar energy by 2010.*
  Sentence: *14. The Czech Republic undertook to achieve an 8% share of production of electricity from*
  Similarity score: 0.72
  Analysis: As mentioned for string-based, the sentence ends at "electricity from" and misses the information that it is renewable sources generally, not specifically solar. What remains is semantically almost the whole fact, so the score is high while the discriminating detail is cut away.

- **Paraphrase detection**

Fact: *Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.*
Sentence: *81. Shortly after Perenco purchased its interests in Blocks 7 and 21, international oil prices began*
Similarity score: 0.70
Analysis: The same problem as the embedding where the sentence stops before showing the words "rise dramatically".

Fact: *The amount of damages awarded to the claimants was reduced by 50%*
Sentence: *end of the day, as was seen in Part XII, the damages awarded to Claimants were reduced*
Similarity score: 0.72
Analysis: The sentence stops at "were reduced," not mentioning "by 50%." Here the overall claim is mentioned but the exact precision is not, yet it receives a high similarity score; In the document the next sentence just mentions "significantly" and no percentage.

- **NLI**

Fact: *INECEL ceased to exist in 1996.*
Sentence: *October 1996, providing inter alia that INECEL be liquidated within a certain time*
Entailment score: 0.99
Analysis: Although the pair shares the core idea, the next sentence in the document mentions "[certain time] period", so not exactly 1996.

Fact: *The investigations into TS involved investigations of TS managers.*
Sentence: *enough evidence to go forward against Taki spravy or its management for crimes of money*
Entailment score: 0.91
Analysis: The pair share the core idea and the model even captured the abbreviation (Taki spravy = TS), however the previous sentence mentions "there was not [enough]" therefore it contradicts the claim.

5.5.2 Chunk-level Evaluation Effect on Large NLI Model

Table 4.6 shows chunk-level candidate comparison improves the macro-F1 score for the large NLI model only, and other approaches either stay the same or decrease slightly. Therefore, at chunk-level, only the large NLI model error analysis was evaluated. This follows how each approach measures textual overlaps. String-based comparison, embedding similarity, and paraphrase detection performs at the surface level comparison, either by character overlap or vector similarity, larger candidates add incidental overlaps that raises scores for both correct and incorrect matches, without improving the discrimination. On the other hand, NLI performs by reasoning over the whole candidate to classify the fact, so a larger candidate can contain the information required to either verify or contradict a fact, which a sentence-level candidate truncated (Section 5.5.1). Of the 36 similar false positives (Section 5.5.3) chunk-level NLI recovers

12 cases. This effect can been seen as some examples mentioned in Section 5.5.1 are correctly classified as Contradicted by the large NLI model at chunk-level.

> ***Fact:*** *The Czech Republic committed to an 8% share of electricity from solar energy by 2010.*
> ***Sentence:*** *14. The Czech Republic undertook to achieve an 8% share of production of electricity from*
> **Captured by Chunk:** "[from] renewable sources of energy ("RES")... by 2010."
> **Analysis:** The fact states that 8% share of electricity is from solar energy, however the chunk shows clearly that it is from renewable sources of energy ("RES"), therefore correctly classifying the fact as Contradicted.

> ***Fact:*** *Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.*
> ***Sentence:*** *81. Shortly after Perenco purchased its interests in Blocks 7 and 21, international oil prices began*
> **Captured by Chunk:** "[began] to rise dramatically"
> **Analysis:** The fact states that after Perenco and Burlington acquired the interests, the oil prices dropped shortly, but the chunk captured the words "rise dramatically", therefore the fact is correctly classified as Contradicted. This part was truncated in the matched sentence.

> ***Fact:*** *TS was the target of a long-running campaign of oppression by international agencies*
> ***Sentence:*** *This proposition is that Taki spravy was the target of a long-running campaign of oppression*
> **Captured by Chunk:** "[oppression] by State agencies"
> **Analysis:** The fact states that the long-running campaign of oppression was done by international agencies, but the sentence was truncated at "oppression" without showing who was in charge of the oppression. The chunk clearly captures the next sentence mentioning "[oppression] by State agencies", therefore the fact is correctly classified as Contradicted.

5.5.3 Shared False Positives

Table 5.1 shows two Contradicted class facts misclassified as Verified by all string-based comparison, embedding similarity, and paraphrase detection approaches. Although these three approaches rely on different scoring mechanisms, they fail on the same facts, indicating the difficulty lies in the fact-candidate pairs themselves, rather than a single approach. In both examples, the pair differs because of a single digit (1980 vs 1990) or a word (convicted vs arrested) and because this is a small change compared to the full string, the similarity scores remain high and the fact is misclassified as Verified.

This is the central limitation shown by the error analysis: on a dataset where Contradicted class facts are minimally edited from otherwise near-identical source sentences, string-based comparison, embedding similarity, and paraphrase detection cannot reliably separate them, regardless of the specific approach, because the small changes barely shift the score. NLI, however, is in principle sensitive to these differences, yet at the sentence-level it also fails on many of the same facts, for different

Method	Threshold	Comparison/Entailment Score
Fact 1: "CEMSA started exporting cigarettes in 1980."		
RapidFuzz `token_sort`	55	64
Embedding	0.65	0.92
Paraphrase detection	0.64	0.79
Fact 2: "Three key Yukos officers were convicted between July and October 2003."		
RapidFuzz `token_sort`	55	65
Embedding	0.65	0.90
Paraphrase detection	0.64	0.90
NLI small model	0.3	0.60
NLI large model sentence-level	0.2	0.30
NLI large model chunk-level	0.95	0.97

Table 5.1: False positives shared by either some or all of the scoring approaches. Both facts have the gold label *Contradicted*; in every row the score exceeds the run's threshold, so each prediction is a false positive.

reasons. One reason is that because the required information often lies outside a single sentence (Section 5.5.1). Therefore the limitation and misclassified facts are shared between approaches, but the cause differs according to the approach.

Similar False Positives (Contradicted facts classified as Verified)	Count
By all string, embedding, and paraphrase	76
By all string, embedding, paraphrase, and both NLI models at sentence-level	36
By all string, embedding, paraphrase, both NLI models and NLI chunk-level	24
By all approaches and IBFD baseline	10

Table 5.2: Shared false positives across approaches. Each row is a subset of the one above: 76 Contradicted facts are missed by all similarity approaches, 36 of these are also missed by both NLI models. Chunk-level NLI recovers 12 of the 36 cases, leaving 24 shared false positives, and 10 of those are missed by the IBFD baseline as well.

Table 5.2 illustrates this overlap. Of the 152 Contradicted facts, 76 cases are misclassified by string-based comparison, embedding similarity, and paraphrase detection, and 36 cases by every sentence-level approach including both NLI models (Appendix Table A.4). Chunk-level large NLI model on its macro-F1 optimal threshold (Section 4.5), recovers 12 of the 36 cases. Of the 24 cases that chunk-level NLI still misses, the IBFD LLM-based system correctly classifies 14 but still misses 10 cases. Therefore, the 10 similar misclassifications between both lighter-weight and LLM-based systems shows that some Contradicted facts are difficult to capture, regardless of the approach. However, the 14 cases mark a capability gap between the lighter-weight approach and LLM-based system. The two examples of facts mentioned in Table 5.1 shows the comparison between the approaches, where fact 2 (Yukos) was misclassified for all approaches, fact 1 (CEMSA) was correctly classified by both NLI models as Contradicted class, while both facts were correctly classified as Contradicted by IBFD baseline.

5.5.4 Irregular or Inconsistent Facts

While looking at the 899 facts inside the dataset, some of them have an irregular structure. Some facts represent something over-general which by itself can lead to being true for candidate and false for another. Also, some spelling differences and truncated facts are visible among these 899 cases; as the spelling differences are not the main source of misclassification, some examples are presented in Appendix A.1.3. Out of the 899 facts, roughly 31 cases fall into the truncated or over-general categories. Although this is a small portion of the dataset, it still affects the Contradicted results as 15 out of 31 are Contradicted class facts. Also, 3 of these 15 cases are among the similar false positive cases missed by all approaches (Section 5.5.3). The following are some examples for the mentioned cases and each fact was checked across all approaches (including large NLI model chunk-level) at their macro-F1 optimal thresholds.

- **Truncated Fact**

  Fact: *9REN began investing in Spain in 20*
  Analysis: The fact is truncated. This format can lead to high scores for many fact-candidate pairs.
  Approaches Performance: Only the large NLI model, on both sentence-level and chunk-level, classified this fact as Contradicted.

- **Over-General Fact**

  Fact: *The case number is ICSID Case No.*
  Analysis: This exact string is represented six times in the dataset, for six different documents. The fact is too general because it does not indicate what the case number actually is, and depending on the context of the related document, many fact-candidate pairs can receive a high score.
  Approaches Performance: Among all the approaches, at least two of these six facts were among the false positive cases. String-based (RapidFuzz) approach was the only approach to classify all six facts correctly.

  Fact: *The number 30 is mentioned in the sentence.*
  Analysis: The fact is also too general because there might be multiple sentences where number 30 is mentioned, especially when there are numerous digits present inside the documents (articles, years, amounts, etc.).
  Approaches Performance: Among all the approaches, only string-based comparison and embedding similarity classify this fact as Contradicted.

  Fact: *The event occurred in May 1998.*
  Analysis: The fact is too general because there is no clarification as to what event is intended and a candidate might result in a high score, as was represented in Section 5.4.
  Approaches Performance: All scoring approaches misclassify this fact as verified, showing that its structure leads to finding a candidate with high similarity/entailment score.

5.6 Summary

This chapter examined the reasons scoring approaches misclassify facts, focusing on the false positive cases. The per-approach analysis showed that the errors are caused by recurring patterns and relate to how each approach compares the similarity between facts and candidates. While facts are paraphrased versions of the document, the Contradicted class facts are typically produced by small edits of the source text – digit change, swapped words, adding or removing negation – therefore many words are similar between the fact and candidate and the small change which flips the fact's meaning does not noticeably lower the similarity scoring. In other cases, the matched candidate makes no verifiable claim at all, such as a heading or a party name that while sharing similar words with the fact, does not assert anything. The synthesis (Section 5.5) showed that these errors are not limited to any one method. The same Contradicted class facts are misclassified by different approaches (Table 5.1), which indicates the difficulty is in the fact-candidate pairs rather than a single approach. How sentences are segmented adds to this, leaving candidates incomplete so that the detail needed to either verify or contradict a fact is truncated or falls in the next or previous sentences the approach never sees. NLI does improve the performance in some cases by capturing hints such as dates and months, and the large model produces fewer unsupported high-confidence errors. However, NLI does not escape the main problem: when the hint for contradiction is outside the matched sentence (Section 5.5.1) and defensible entailment on a non-source candidate (Section 5.4), no sentence-level approach can capture it completely.

Overall, these findings indicate that the limitation on this dataset is not the choice of any single scoring approach (string-based comparison, embedding similarity, paraphrase detection, or NLI), but partly the structure of the data and partly the capability gap between the lighter-weight approach and an LLM (Section 5.5.3). The contradicted facts remain lexically close to their sources, and the deciding detail is often outside the single sentence being scored, which is largely a segmentation problem, cutting candidates before the required information. These issues are further discussed in Chapter 6.

Chapter 6

Discussion and Conclusion

This chapter discusses the findings of the thesis, answers the research questions, and examines the limitations of the present study. Moreover, it puts forward directions for future work according to the limitations.

6.1 Interpretation of Findings

The clearest pattern across the results is that the approaches utilized in this thesis perform within a narrow band. Whether string-based similarity, embedding-based similarity, or paraphrase detection, the approach reaches a macro-F1 score of roughly 0.55 to 0.57, regardless of how differently each approach evaluates how similar a fact-candidate pair is. A paraphrase-tuned bi-encoder does not improve on a general embedding model, and neither improves on character-level edit distance. This convergence suggests that the limiting factor on this dataset is not only how similarity is examined between a fact and a candidate, but also the task itself. The Contradicted-class facts, in addition to being paraphrased, are mostly produced by small edits to a source sentence such as a changed digit or added negation, so the fact and its source remain highly similar on the surface. Because the change that flips the fact from true to false is small relative to the shared text, the approaches cannot separate them. The error analysis (Chapter 5) showed this directly: the same Contradicted-class facts are misclassified by all approaches (Section 5.5.3). Also, some facts were observed with irregular structure (too general or truncated), making them challenging to correctly classify for any scoring approach. These findings suggest that the gap is based on both task difficulty and a capability difference between the lighter-weight approach and the LLM.

The two granularities used for comparison performed differently for different approaches. For the string-based, embedding-based, and paraphrase detection, moving from sentence-level to chunk-level candidates makes little difference. However, the NLI cross-encoder behaves differently, and only the large model benefits. Because of the joint comparison architecture of the cross-encoder, it can use the surrounding context to judge relationships that span more than one sentence. At the sentence-level, this context is often missing, and models fail to see the required hints causing them to underperform. Part of this failure is caused because of the pre-processing: OCR and sentence segmentation often truncated sentences before the information required for classification appears, which is why extending the candidates to chunks resolves some of the misclassifications for the large NLI model. At the chunk-level, the larger NLI model improves noticeably, seeing an increase in macro-F1 score from 0.56 to 0.64,

the highest score in this study. The smaller model does not improve, which suggests that the benefit comes from the larger model's capacity. The different performance according to the chosen granularity indicates that the right candidate granularity may be different according to the approach.

For IBFD, the findings offer a cost-aware choice. The lighter-weight approach reaches the highest macro-F1 score of 0.64, without the API cost of an LLM-based system (GPT-5.4 mini for IBFD). The IBFD's case law collection extends well beyond the 14 award documents examined in this thesis. On a larger scale, the repeated runs required for prompt refinement and ongoing production use would likely lead to a meaningful cost to the company. Moreover, unlike the LLM-based system, the lighter-weight approach can be run entirely locally, meaning the confidential case law data would never need to leave IBFD's infrastructure. Where the computational budget and confidentiality of the data are the binding constraints, the lighter-weight approach is worth considering and can narrow the gap to LLM performance while avoiding external data transfer.

6.1.1 Research Questions

The three sub-questions ask about how each similarity scoring approach performs in this study. For the string-based similarity (Sub-question 1), both libraries reach the macro-F1 score of 0.56 against the baseline's 0.75, with the two libraries performing almost identically. For the embedding-based similarity (Sub-question 2), dense embeddings do not improve on string-based or paraphrase detection. For the more semantically-informed approaches (Sub-question 3), paraphrase detection almost performs like a general embedding model. NLI is the only approach to move the Contradicted class scores noticeably, and reaches the macro-F1 score of 0.64 against the baseline's 0.75 macro-F1 score at the chunk-level. The main research question asked about the performance of the lighter-weight approach compared to the IBFD LLM-based system. On this dataset, no lighter-weight pipeline matches the baseline's macro-F1 score of 0.75, but the best configuration — chunk-level large NLI model — manages to close the gap, reaching a macro-F1 score of 0.64. Therefore, the lighter-weight approach can get close but not match an LLM-based system for fact-verification in case law documents.

6.2 Recommendation For IBFD

For IBFD, the findings can suggest a conditional recommendation rather than a single answer. If the priority is to achieve the highest possible macro-F1 score, the lighter-weight approach is not yet a replacement for the LLM-based system, since there is a gap in performance, particularly on Contradicted class facts. But if the priority is reducing the computational and API cost while accepting some loss in performance, the lighter-weight approach can be an applicable option. Also, considering the subject matter experts (SMEs) at IBFD, having a low Contradicted class recall is problematic. A contradicted fact labeled Verified is an unverified error passed to the expert, which is against the purpose of a fact verification system; capturing incorrect facts. If the Contradicted recall is low, an SME would need to double check the facts, shifting the task back to the manual checking it aimed to remove. Therefore, an applicable approach needs to have both an acceptable macro-F1 score and Contradicted class performance, which the current lighter-weight approach lacks.

6.3 Limitations

The current task is faced with limitations in three main areas: the labeling and evaluation setup, the lighter-weight approach design, and the data. The first is the labeling. The three-class gold labels were collapsed to binary by merging the 20 Insufficient Evidence facts into the Contradicted class, to align with the binary scoring approaches. This simplifies the original labeling, and the results should be examined as binary classification (the three-class baseline results are reported in Appendix (Table A.2). The granularity is limiting here as well. IBFD assigned each gold label by comparing a fact against chunks of sentences in the award documents. While this thesis examines a chunk-level comparison as a second method, the original IBFD chunking strategy was not available. Therefore, the chunk boundary selection was not fully optimized according to each document's structure. Because the chunk-level comparison helped NLI obtain a higher macro-F1 score, more carefully designed chunks can further improve the results.

A second group of limitations concerns how the lighter-weight approach is designed. The designed system intentionally used small, off-the-shelf bi-encoders without domain-specific fine-tuning. Therefore, it is not possible to say what effect a larger encoder or fine-tuning on the legal text would have on the results. Moreover, each scoring approach was evaluated independently, so the study does not report the performance of a combined system. Whether combining them would improve the results or not, was not tested in this study. Also, thresholds were examined and selected on the same dataset for evaluation, with no separate data performing as a development set.

A third group of limitations concerns the data. The dataset contains 14 case law award documents and 899 atomic facts, all from IBFD. This is suitable for an exploratory study, but limits the generalizability of the results and conclusions to other case law repositories, languages, or legal documents from different jurisdictions. Both the performance and ordering of the four scoring approaches could differ on another dataset. In addition, some of the facts (roughly 31 out of 899) have irregular design, either truncated or over-general, and make it difficult for scoring approaches to correctly classify them (Section 5.5.4). As 15 of the facts have the gold label of Contradicted, the performance of the approaches on these cases is slightly understated compared to well-designed facts. Finally, the award documents were processed with Azure Document Intelligence. Because this method operates as a fully managed endpoint and does not provide control over the process, the extraction functions as a black box. This text extraction introduces noise and occasional sentence-segmentation boundary errors, which affects the procedure downstream. The actual rate and distribution of OCR errors for this dataset were not measured in this thesis.

6.4 Future Work

The limitations discussed in the previous section lay the groundwork for several directions for future research related to this study. The first is to combine the scoring approaches rather than evaluating them independently. The error analysis (Chapter 5) showed that the approaches misclassify different facts, where for example, NLI resolves the date misclassifications that the string-based approach makes, and the large NLI model, at the chunk-level, captures more of the Contradicted class than other approaches. Two designs are worth examining for this task. The first is an aggregation

step that merges the approaches' predictions, for instance through weighted voting. The second is a cascade design in which the cheaper and faster approaches act as an early filter and the more computationally expensive NLI model is then applied to the remaining harder cases. The prior work on legal case matching illustrates that combining representation-based and interaction-based signals improves the performance in legal case matching (Fan et al., 2024), which is consistent with this direction.

The second direction concerns the paraphrase detection approach. In this thesis, a bi-encoder model fine-tuned on paraphrase pairs was used for paraphrase detection approach. This model produced a continuous score using cosine similarity, not as a binary paraphrase classification; the fact is the paraphrased version of the candidate or not. A model, like a cross-encoder, trained to provide a paraphrase decision output as whether one fact is the paraphrase version of a candidate or not, may capture the fact-candidate relationship differently and also be compared against the NLI cross-encoder performance.

The third direction is to expand the dataset. In the short term, expanding the dataset can happen inside IBFD by adding more case law documents or by creating more contradicted facts, which would give balanced per-class estimates and would make supervised fine-tuning of the embedding and paraphrase detection models possible. In the longer term, evaluation on case law from other jurisdictions or languages included inside IBFD's collection would assess the performance of the current pipeline beyond the present setting.

A fourth direction concerns the text extraction. The OCR output faced noise and segmentation problems, where information like charts and tables were scrambled and footnotes and page numbers were intertwined with the text. Another direction for future work is to characterise OCR output explicitly and manually compare the extracted sentences against the source award documents, measure the rate and type of extraction and segmentation errors, and solve the errors to have a better output. Also, comparing different extraction methods from PDFs can be valuable and lead to a better text output. These actions can affect the results since these errors propagate into every approach downstream. Also, the chunk-level results show that expanding candidates improves the results, therefore larger-sized chunks might improve NLI results further. This is consistent with findings of Laban et al. (2022), who illustrated that adjusting text granularity and size of the context can improve NLI performance. Evaluating this requires designing new chunks and re-running the cross-encoder for the large NLI model, which was not feasible within the internship duration and data-access constraints, so it is left to future work. Also, as the chunks used by IBFD were not available, performing the chunk-level comparison on the same chunks as the LLM-based system can provide a more accurate comparison.

6.5 Conclusion

In closing, this thesis showed how well a lighter-weight pipeline can perform fact verification on case law documents, compared against an LLM-based system. The lighter-weight approach did not match the baseline's performance (macro-F1 score 0.75), but at its best configuration, chunk-level scoring with the large NLI model, the model reached the macro-F1 score of 0.64, without the API cost required for an LLM-based system. Its main finding is that, on this dataset, the difficulty partly lies in the structure of the task – paraphrased Contradicted class facts that differ slightly from the source

– and partly in a capability gap between the lighter-weight approach and the LLM, rather than in the choice of the scoring approaches. This pattern was consistent across every approach where 24 Contradicted facts were misclassified by every lighter-weight approach, including the NLI chunk-level, and even the LLM-based system struggled on 10 of these cases, indicating the limitation is partly the property of the data rather than of any single system.

Appendix A

Supplementary Data and Error Analysis

Case Name (Shortened)	Number of Unique Words
Yukos Universal v. Russia	17,500
Perenco v. Ecuador	14,600
Lone Star v. Korea	11,800
Burlington v. Ecuador	11,100
Ampal v. Egypt	7,000
Manolium-Processing v. Belarus	6,600
Paushok v. Mongolia	6,500
Mamidoil v. Albania	6,300
9REN Holding v. Spain	6,200
JSW Solar v. Czech Republic	4,900
Duke Energy v. Ecuador	4,600
Tokios Tokelés v. Ukraine	4,000
Feldman v. Mexico	3,600
Enron v. Argentina	1,600

Table A.1: Approximate Number of Unique Words per Case File

Table A.1 reports the number of unique words per award document, after removing standard function words. The distribution illustrates the lexical diversity and complexity of the source texts, which scales relatively with the increase in document length. For the lighter-weight approaches used in this thesis, a larger vocabulary directly affects the difficulty of capturing the similarities matching: it raises the likelihood that a paraphrased fact uses entirely different terminology (synonyms) from its source sentence, which surface-level approaches cannot capture.

Table A.2 illustrates the IBFD LLM-based system performance when evaluating all three classes.

Class	Precision	Recall	F1-Score	Support
Verified	0.940	0.857	0.896	747
Contradicted	0.829	0.477	0.606	132
Insufficient evidence	0.106	0.750	0.185	20
Accuracy			0.799	899
Macro Avg	0.625	0.695	0.562	899
Weighted Avg	0.905	0.799	0.838	899

Table A.2: IBFD LLM-Based system performance on the full three-class evaluation set

A.1 Detailed Error Analysis

A.1.1 NLI Performance

The following six facts occurred as false positives exclusively within the Natural Language Inference (NLI) scoring approach, meaning they were misclassified by the NLI model but correctly handled by the string-based comparison, embedding similarity, and paraphrase detection approaches:

1. Law Nr. 30 from 1998 provided that goods imported into Ecuador were entitled to duty exemptions.

2. Mamidoil labelled its claim for reimbursement of allegedly overpaid tax duties as a claim for unjust enrichment under the Civil Procedure Code.

3. The claimants argued that the assessment of VAT and corporate taxes on various operations of the TGS was illegal under Argentinian law.

4. The Mexican President revised the IEPS law effective 1 January 1992.

5. The Tribunal found that the revocation of the tax-exempt status granted to EMG until 2025 was a measure tantamount to a breach of the FET standard.

6. The Tribunal ruled that the damages for the revocation of the EMG licence must be calculated beyond 2025.

A.1.2 NLI small model over-confidence entailment scoring

A.1.3 Spelling Mistakes Among the 899 Atomic Facts

These examples show that some facts have different spellings from the text inside the source document: the bold words inside each fact represent this. Some of these spelling mistakes are repeated inside several facts: licence 7 times, programme 4 times, and authorised 4 times. These cases are differences between American English and British spelling, which the LLM altered while generating the facts.

- ***Fact:*** *The Tribunal ruled that the damages for the revocation of the EMG* ***licence*** *must be calculated beyond 2025.*
  Analysis: The text inside the source document has the word as "license".

Fact	Matched Sentence	Entailment Score
1. The state-owned companies were in the oil sector	WASHINGTON, D.C.	0.97
2. The claimants made investments in the oil industry of Argentina	WASHINGTON, D.C.	0.83
3. Several Southern Argentine provinces assessed VAT and corporate taxes on various operations of the TGS	WASHINGTON, D.C.	0.91
4. The tax procedures initiated by Perenco concerned unpaid amounts.	WASHINGTON, D.C.	0.74
5. Perenco initiated tax procedures in 2009.	WASHINGTON, D.C.	0.64

Table A.3: Entailment scores for facts matched with the same sentence "WASHINGTON, D.C.", showing the small NLI model performance in giving high entailment score to a fact-sentence pair where there is no clear entailment between them.

- ***Fact:*** *The tax authority* ***authorised*** *new rebates in December 1997.*
  Analysis: The text inside the source document has the word as "authorized".
- ***Fact:*** *Argentina began a privatization* ***programme****.*
  Analysis: The text inside the source document has the word as "program".

A.1.4 Large NLI model at chunk-level and LLM Performance on the Shared False Positive

Fact	Large NLI Chunk-Level Prediction	LLM Prediction
1. A claim for denial of justice can be considered as an appeal against decisions of the national judiciary.	Verified	Contradicted
2. According to Ecuador, the price increase helped the negotiation of PSCs.	Contradicted	Contradicted
3. Ecuador committed to reimburse customs duties.	Verified	Contradicted
4. Ecuador rejected Perenco's proposal to transfer interest payments to a savings account	Verified	Verified
5. Law 24 amended the Hydrocarbons Law.	Contradicted	Contradicted
6. Law 42 with a 99% tax rate did not substantially deprive Burlington of the value of its shares.	Verified	Contradicted
7. Mamidoil (Claimant) is an Albanian corporation	Verified	Contradicted
8. Mamidoil's core business is the acquisition of petroleum products	Verified	Contradicted
9. Mexico imposed an excise tax on the production and sale of cigarettes under the Impuesto Especial Sobre Servicios (IEPS) law	Verified	Verified
10. Oil prices dropped shortly after Perenco and Burlington acquired the interests in Blocks 7 and 21.	Contradicted	Contradicted
11. One of the measures is Law 42 (ii) with 90%.	Verified	Contradicted

Continued on next page...

Fact	Large NLI Chunk-Level Prediction	LLM Prediction
12. Perenco Ecuador Limited (Perenco) is a company incorporated in Iran	Contradicted	Contradicted
13. Perenco Ecuador Limited (Perenco) was engaged in tobacco exploration and exploitation in Ecuador	Contradicted	Contradicted
14. Perenco initiated a process for early termination of contracts.	Verified	Verified
15. Perenco initiated tax procedures in 2009.	Contradicted	Verified
16. Perenco is a Persian oil company controlled by French nationals	Contradicted	Contradicted
17. Petroecuador bid the crude production from Blocks 7 and 21 as a result of the enforcement measures.	Contradicted	Verified
18. Revenues from the TVPEE were directed to the State General Budget.	Verified	Verified
19. The amount of damages awarded to the claimants was reduced by 50%	Verified	Contradicted
20. The claimants in this case are three companies.	Verified	Contradicted
21. The Czech Republic withdrew the tax incentives in 2009	Contradicted	Contradicted
22. The expropriation claim did not proceed	Verified	Contradicted
23. The feed-in tariff (FIT) would be set by the Czech Electricity Regulatory Office (ERO).	Contradicted	Verified
24. The four tax reassessments issued by the Tax Ministry were for the tax years 2000 to 2004.	Verified	Contradicted
25. The participation contracts included a tax stability clause.	Verified	Contradicted
26. The price of gold as of the sale date would determine the market price.	Verified	Verified
27. The purpose of the rental agreement was to establish a fuel storage centre	Verified	Verified
28. The rental agreement was for using a land parcel at the port of Durres	Verified	Verified
29. The tax authorities alleged that TS was involved in tax fraud	Verified	Verified
30. The tax authorities initiated investigations in March 2002.	Verified	Contradicted
31. The tax procedures initiated by Perenco concerned unpaid amounts.	Verified	Verified
32. The tribunal awarded compensation only for the losses suffered as a result of the sale of Burlington's investments (Blocks 7 and 21).	Verified	Contradicted
33. There were no bids at the auctions held by Ecuador for the seized crude oil.	Verified	Contradicted
34. Three key Yukos officers were convicted between July and October 2003.	Contradicted	Contradicted
35. Yukos emerged as one of the largest oil companies in Russia	Verified	Verified
36. Yukos was incorporated in 1999.	Contradicted	Contradicted

Table A.4: Comparison of the large NLI model at chunk-level and LLM predictions for the 36 similar false positive cases (incorrectly labeled as Verified) between string-based comparison, embedding similarity, paraphrase detection, and NLI at the sentence-level.

Appendix B

LLM Instruction for IBFD LLM-based System

```
GROUND-TRUTH EVIDENCE:
Each chunk is split into sentences marked with [S1], [S2], etc. Use these markers to
↪ reference specific evidence.

{context}

TASK REQUIREMENTS (follow all steps exactly):

1. Evaluate whether the provided evidence supports, contradicts, or fails to verify the
↪  fact.
   - Your evaluation must rely solely on the supplied ground-truth documents.
   - Be critical in your evaluation. Keep a close eye on dates, entity names,
↪ ambiguities.
   - Treat linguistic differences and qualifiers (e.g., "directly", "indirectly", "
↪ majority", "wholly", date ranges, modalities such as "may" vs "does") as materially
↪ important. Do not assume equivalence between different qualifiers or paraphrases.
   - "Verified" requires an explicit, literal match between the fact's claim and the
↪ text of the evidence. Similar, weaker, broader, or paraphrased wording in the
↪ evidence is **not** sufficient for "verified".
   - If the fact asserts a stronger or more specific claim than the evidence supports,
↪ then:
     - If the evidence explicitly contradicts the stronger claim, label the fact "
↪ contradicted".
     - If the evidence does not explicitly contradict but does not provide the required
↪  specific wording, label the fact "insufficient_evidence".
   - Do not infer unstated facts from context, nor chain together multiple inference
↪ steps to create a verification. Use only literal statements present in the provided
↪ documents.

2. Output one verification status, exactly as one of the following strings:
   - "verified"
   - "contradicted"
   - "insufficient_evidence"

3. Provide a confidence score between 0.0 and 1.0, using these definitions:
   - 0.0-0.2: Uncertain / no clear basis
   - 0.2-0.4: Weakly informed
   - 0.4-0.6: Moderately confident
   - 0.6-0.8: Confident
   - 0.8-1.0: Highly confident
```

```
   This confidence reflects your certainty in your *verification judgment*, not the
↪ strength of the evidence itself.

4. Explain your reasoning step-by-step.
   - **Generate the full explanation before emitting the status.** The explanation
↪ should record your literal comparisons, identify any mismatches between the fact
↪ wording and evidence wording, and quote the exact phrases or sentences in the
↪ documents that you rely on (include document name and page number with each quoted
↪ phrase).
   - Cite specific document names and page numbers.
   - NEVER mention chunk ids in your explanation or reasoning step.
   - The explanation must show how the evidence supports your final status, and must
↪ explicitly call out any qualifier or wording differences that influenced the
↪ decision.

5. List the exact chunk_id values you relied on.
   - Chunk IDs are UUID4 strings.
   - Provide the exact IDs, not their indices.
   - You must use at least one chunk_id.

6. List the specific sentence IDs (evidence_ids) you relied on.
   - Use the format "chunk_id:S1", "chunk_id:S2", etc.
   - Reference the [S1], [S2] markers from each chunk.
   - Include ALL sentences that support your judgment.
   - Example: ["abc-123:S1", "abc-123:S3", "def-456:S2"]

7. Use only evidence from the ground-truth documents. Do not rely on outside knowledge.

8. Be precise, objective, and avoid speculation.
   - If a literal reading of the evidence leaves any required element of the fact
↪ unsupported, choose "insufficient_evidence" rather than stretching to fit "verified"
↪ .
   - If the evidence contains an explicit denial of the fact, choose "contradicted".

Be precise and objective. Follow the literal text of the evidence; do not infer,
↪ generalize, or equate non-identical wording.
```

Listing B.1: IBFD LLM-based system verification prompt, reproduced as provided by IBFD. The {`context`} placeholder is populated at run time with the retrieved candidate chunks.

```
## Splitting Dimensions
Split the target sentence along these dimensions - each dimension should typically
↪ produce a separate fact:
- **Temporal** (when): dates, years, time periods
- **Geographic** (where): locations, regions, countries
- **Entity** (who): companies, people, organizations
- **Action** (what): events, actions, changes
- **Manner** (how): methods, approaches, means
- **Relationships**: connections between entities (e.g., "X is associated with Y")

## Example
Target: "In March 2020, the company relocated its headquarters to Austin, Texas, citing
↪  lower operating costs and a more favorable business environment."

Output facts:
1. "The company relocated its headquarters in March 2020"
2. "The company relocated its headquarters to Austin, Texas"
3. "The company cited lower operating costs as a reason for relocating"
```

```
4. "The company cited a more favorable business environment as a reason for relocating"

## Instructions

1. **Extract from TARGET SENTENCE only**: Use the context to understand references (
↪ pronouns, abbreviations, entity names) but do NOT extract facts from the context
↪ sections.
2. **One verifiable claim per fact**: Each fact should assert exactly ONE thing that
↪ can be independently checked.
3. **Split compound statements**: "A and B" should become two separate facts.
4. **Resolve references using context**: If the target sentence uses "it", "the company
↪ ", "they", etc., use the context to identify the actual entity and include that in
↪ the fact.
5. **Repeat the subject**: Each fact must include its subject so it can stand alone.
6. **Preserve specificity**: Do not lose details when splitting - dates, numbers, names
↪  must appear in relevant facts.
7. For each extracted fact, provide TWO parallel lists with a strict 1:1 correspondence
↪ :
    - verbatim_excerpts: The EXACT verbatim text excerpt from the TARGET SENTENCE that
↪ contains this fact. Copy-paste character-for-character from the target sentence only
↪ .
    - facts: The atomic factual statement derived from that excerpt.

   **CRITICAL**: Both lists must have the SAME length. Each verbatim_excerpts[i] must
↪ correspond to facts[i].

8. Avoid opinions, interpretations, or unverifiable claims.
9. If the target sentence contains no extractable factual statements, return empty
↪ lists for both fields.
```

Listing B.2: IBFD atomic-fact extraction prompt, reproduced as provided by IBFD. Applied to each target sentence to generate the atomic facts.

Appendix C

Confusion Matrices

This section illustrates the confusion matrices for threshold sweep used for each scoring approach. The confusion matrices for the chunk-level are presented for the large NLI model, showing both scoring approaches of early-exist and asymmetric.

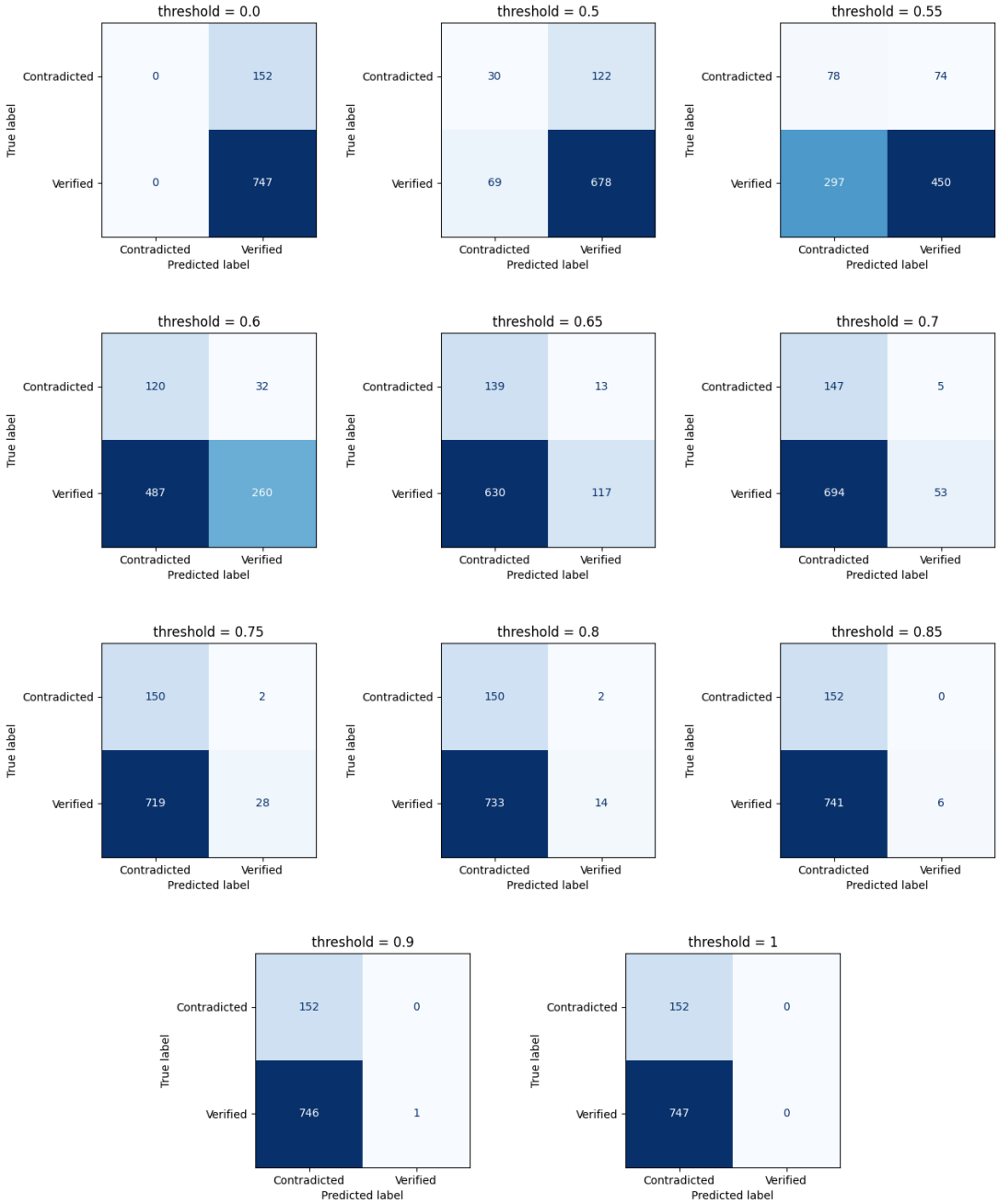


Figure C.1: Confusion matrices for the string-based approach (Levenshtein) across different thresholds.

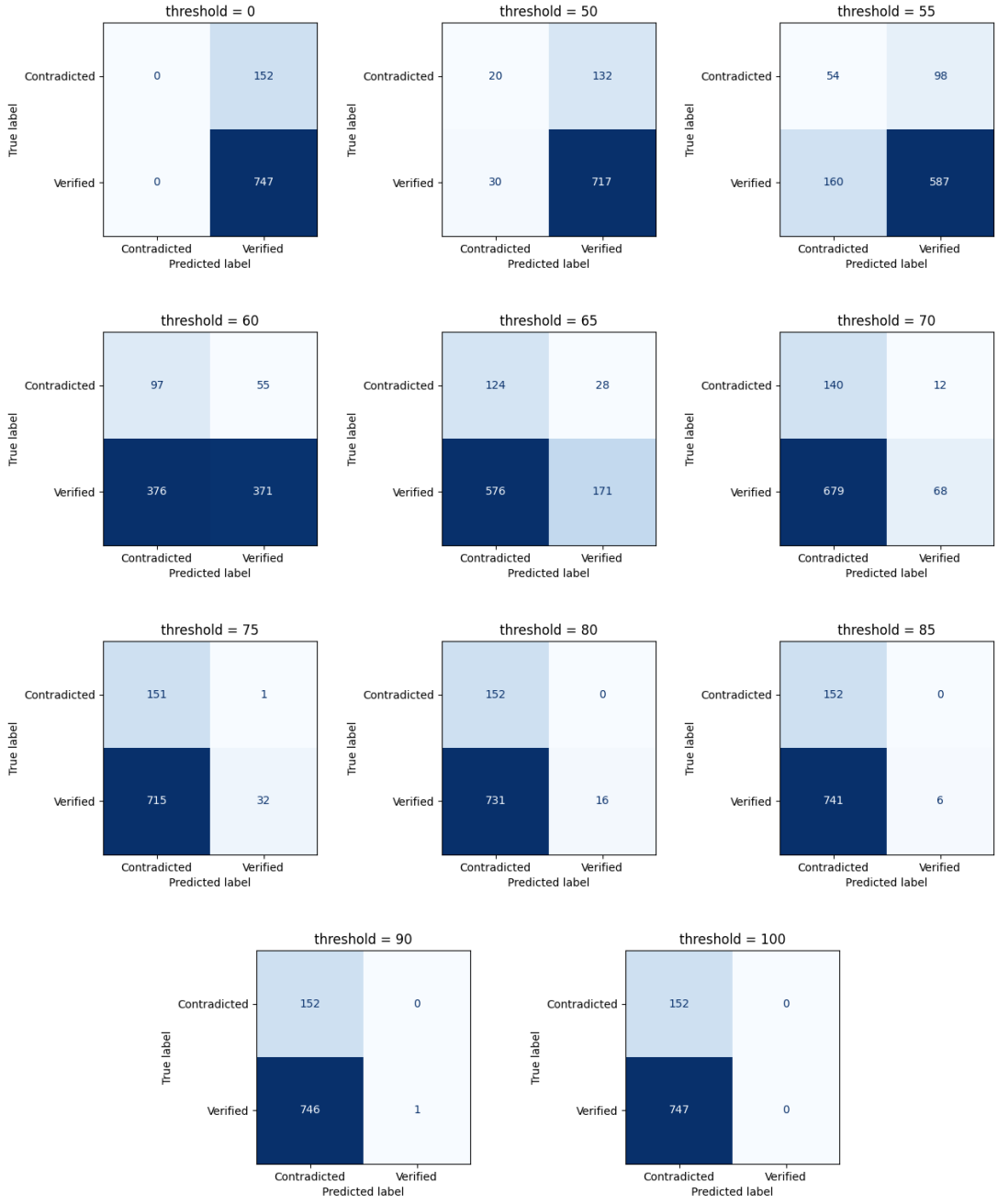


Figure C.2: Confusion matrices for the string-based approach (RapidFuzz) across different thresholds.

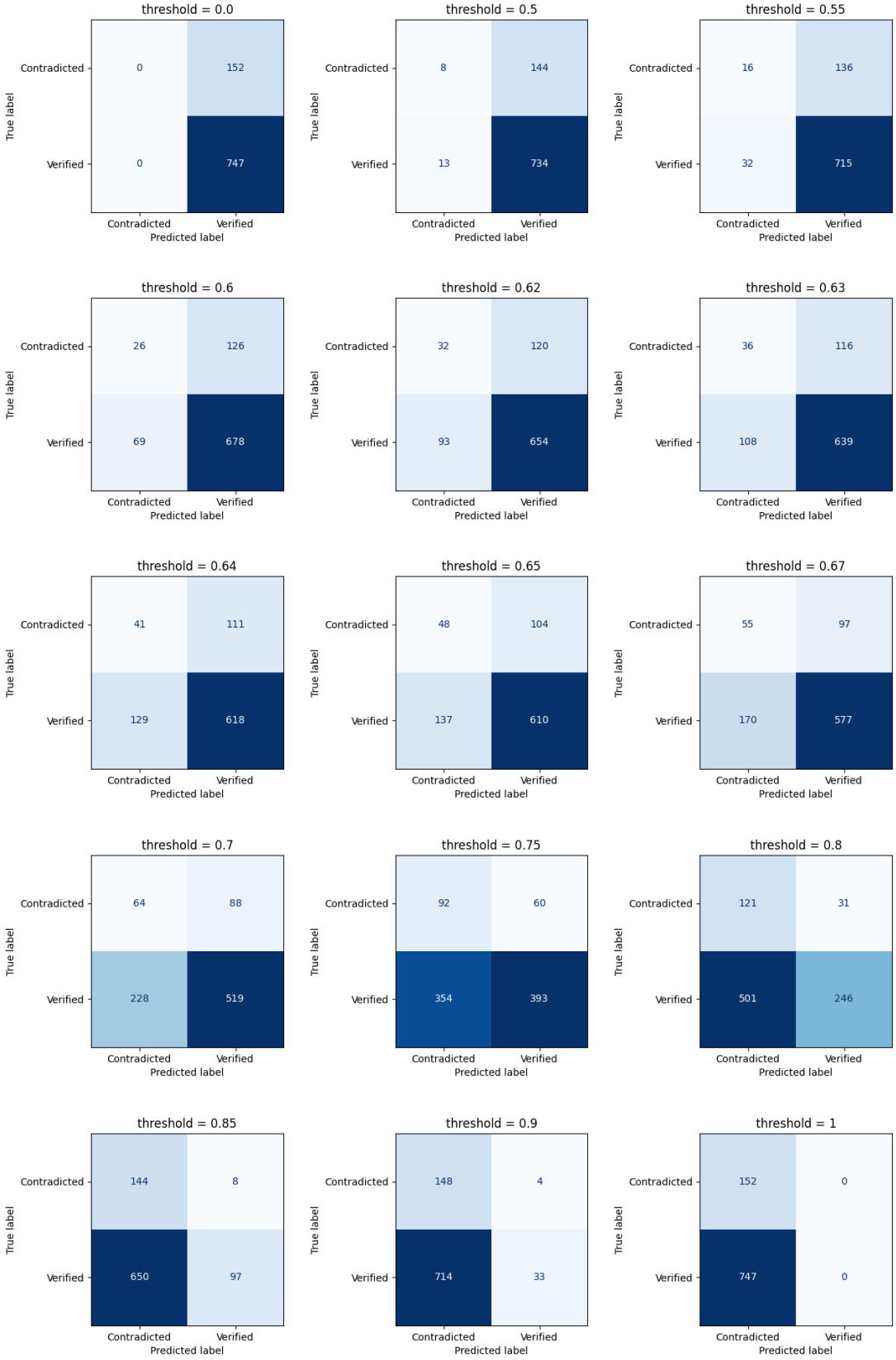


Figure C.3: Confusion matrices for the Embedding-based approach across different thresholds.

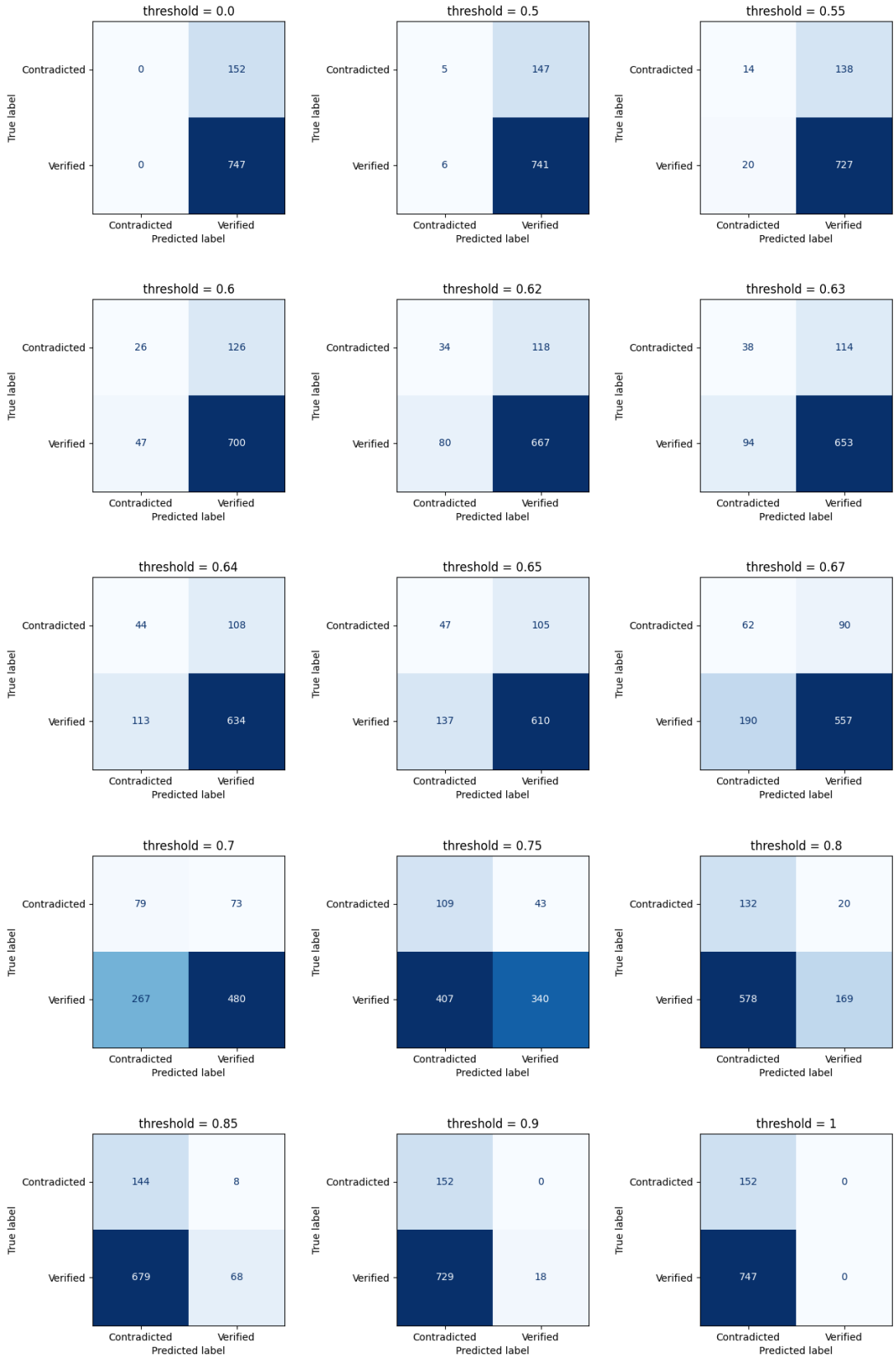


Figure C.4: Confusion matrices for the Paraphrase detection approach across different thresholds.

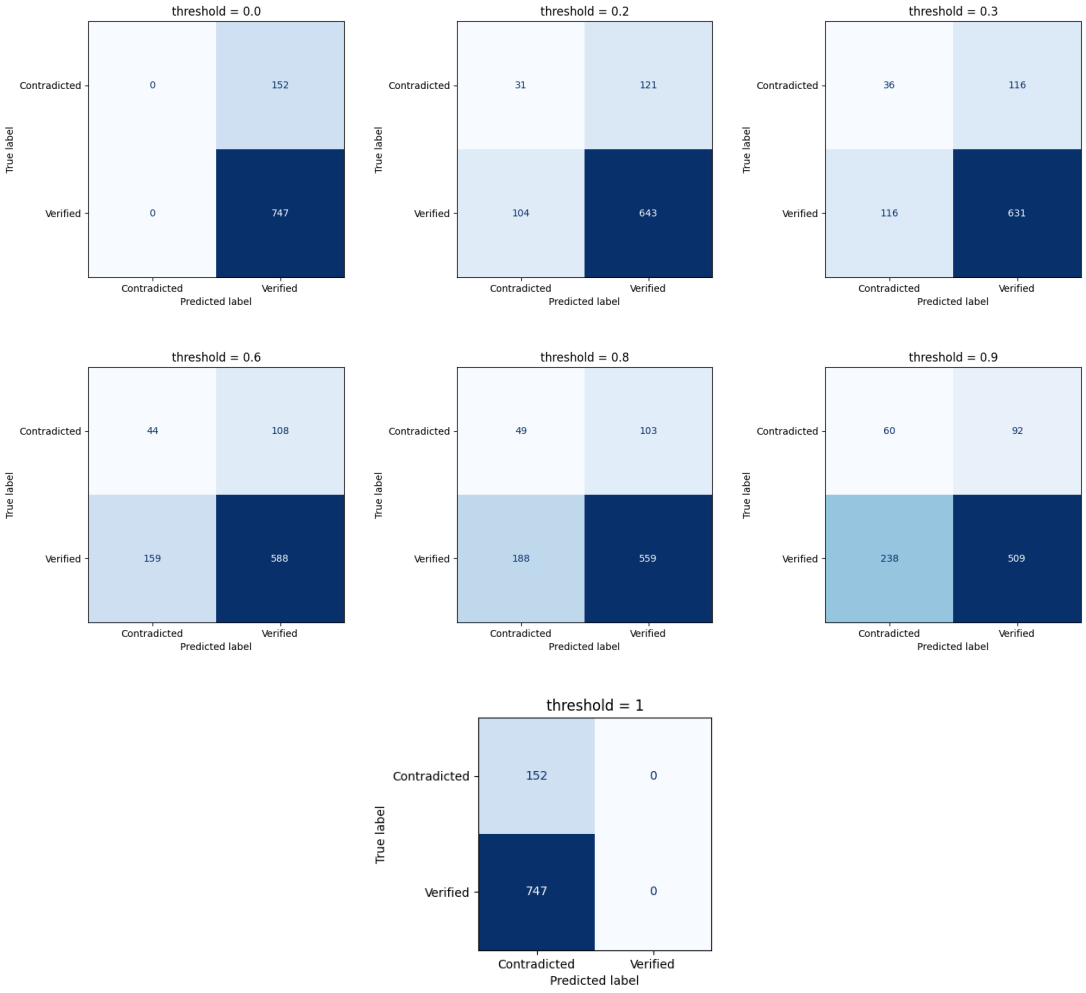


Figure C.5: Confusion matrices for the NLI approach for small model at sentence-level across different thresholds.

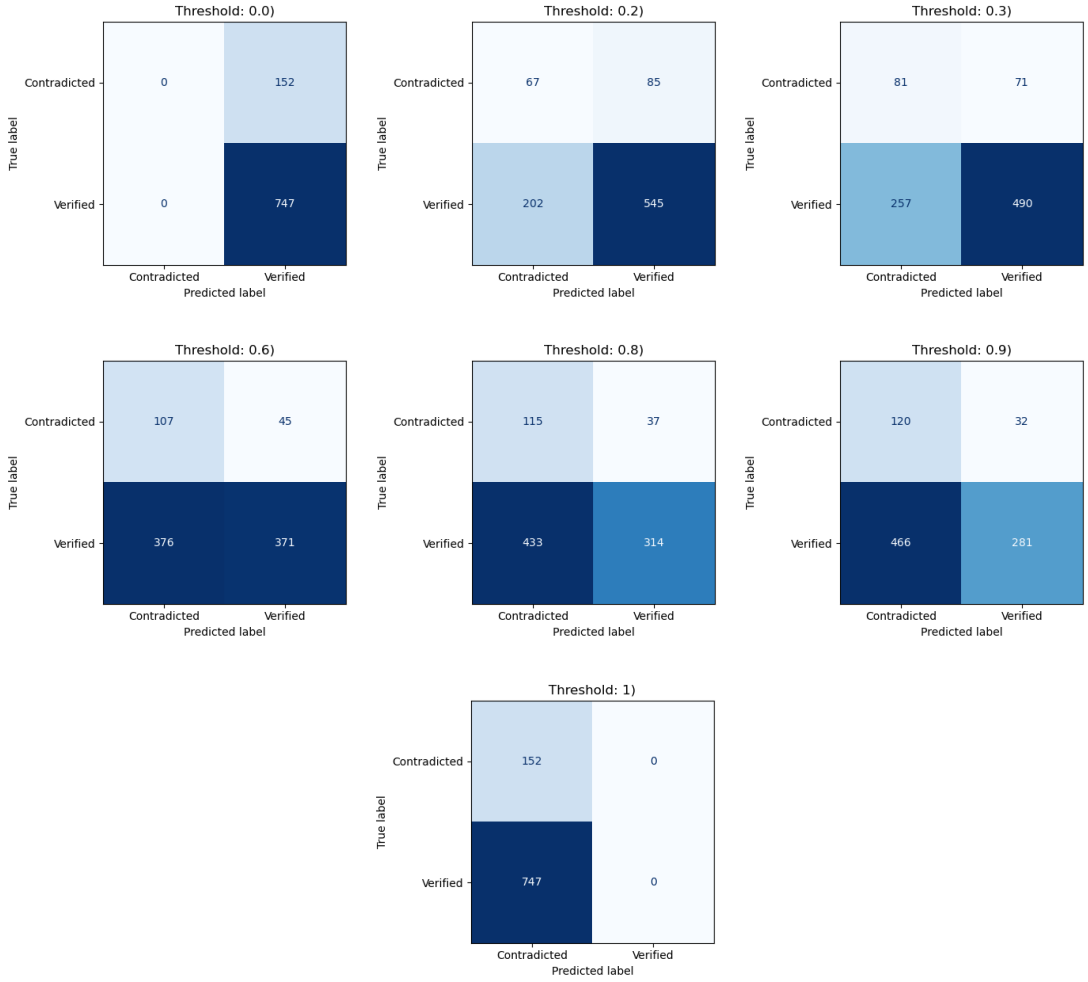
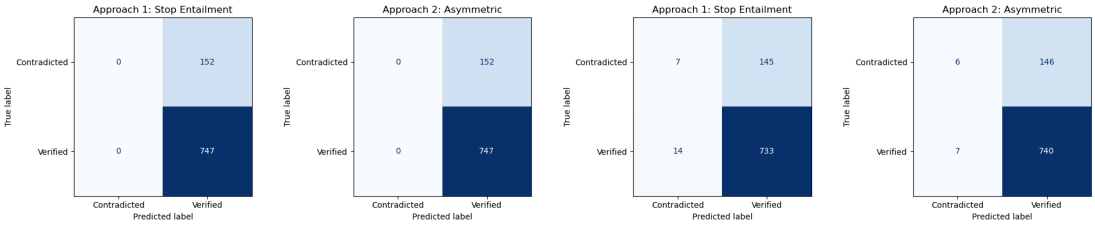
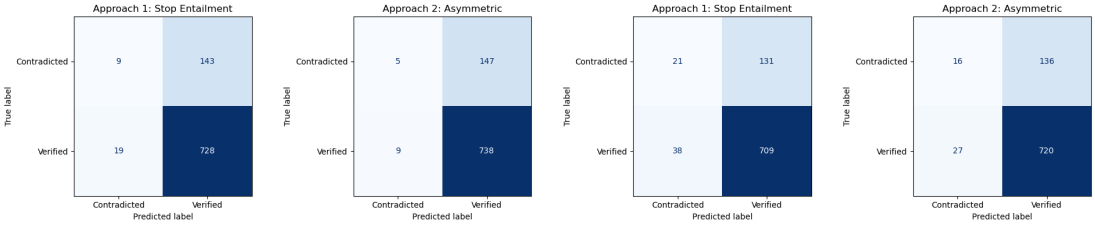


Figure C.6: Confusion matrices for the NLI approach for large model at sentence-level across different thresholds.



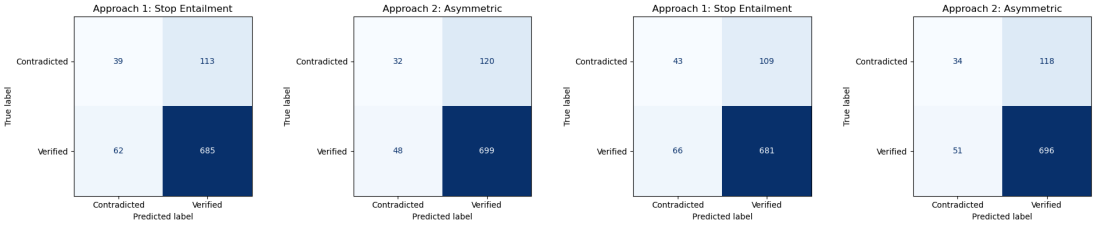
(a) Threshold = 0

(b) Threshold = 0.1



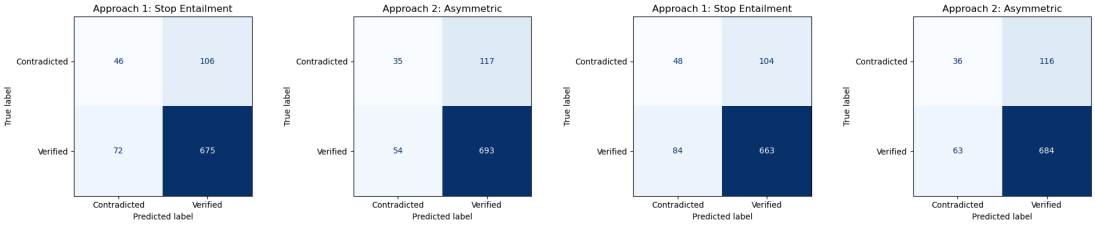
(c) Threshold = 0.3

(d) Threshold = 0.4



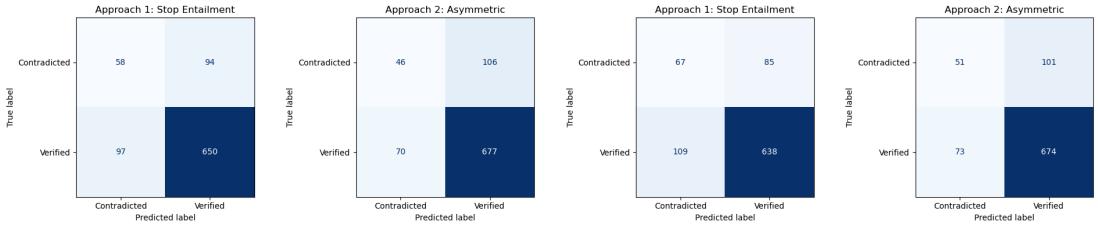
(e) Threshold = 0.5

(f) Threshold = 0.6



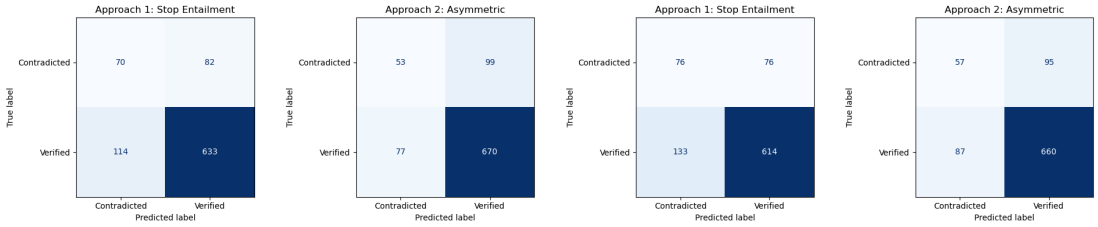
(g) Threshold = 0.7

(h) Threshold = 0.8



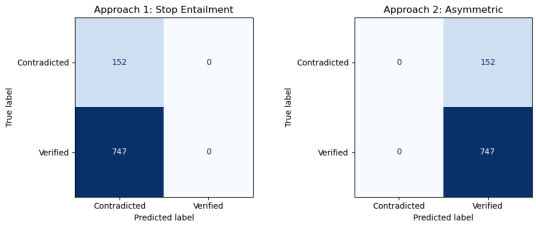
(i) Threshold = 0.9

(j) Threshold = 0.94



(k) Threshold = 0.95

(l) Threshold = 0.97



(m) Threshold = 1

Figure C.7: Confusion matrices for the NLI approach for large model at chunk-level for both scoring approaches across different thresholds. In this figure only, the Early-Exit scoring is labeled as "Stop Entailment".

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